A Formally Verified Pandrosion–Steffensen Scheme

Closed-form real-line analysis and complex multi-start extension
for the equation $z^p = x$, with a Lean 4 / Mathlib v4.7.0 development

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Abstract

This paper combines, in a single document, the real-line analysis of Pandrosion's classical geometric construction for p-th roots and its formally verified complex-multi-start extension.

Part I (real line). We revisit Pandrosion's construction reported by Pappus for approximating $\sqrt[3]{2}$ and generalize it to the computation of p-th roots of any positive real x by inserting proportional means via Thales' theorem. After normalization the iteration reduces to the universal form $s_{n+1} = 1 - (x-1)/(x\,S_p(s_n))$ with $S_p(s) = 1 + s + \cdots + s^{p-1}$, fixed point $s^* = x^{-1/p}$, and readout $v_n = x\,s_n^{p-1} \to x^{1/p}$. We derive the closed form

$$\lambda_{p,x} = \frac{(\alpha-1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1}\alpha^k}, \qquad \alpha = x^{1/p},$$

for the linear contraction ratio, prove it is monotone in both p and the root parameter in the precise senses of Theorem 9.1 (via the closed-form identities $P_p(\alpha) = p\,D_p'(\alpha)$ and $B_p(\alpha) = \sum_{j=1}^{p} j\alpha^{j-1}$, both with all positive coefficients), with explicit large-p limit $1 - \ln(x)/(x-1)$, give non-asymptotic error bounds, extend the theory to $x < 1$ with an explicit threshold α_p^* below which the iteration diverges, and add two accelerations: a canonical starting point $s_0 = h(1) = 1 - (x-1)/(xp)$, and Aitken–Steffensen Δ^2 yielding quadratic convergence with a much smaller constant than Newton. A scaling preconditioning $x = A \cdot x'$ exploits the monotonicity of λ to collapse iteration counts for large x.

Part II (complex extension, formally verified). We extend the iterator $h_{p,x}$ to $\mathbb{C}$, where it has p attractive fixed points — the p-th roots of $1/x$ — separated by a non-trivial Julia structure. A multi-start scheme $\mathrm{seed}_{\alpha,\varepsilon}(z) = \alpha + \varepsilon(z-\alpha)$ with target-dependent ε places any $z \in \mathbb{C}$ inside the quadratic-contraction disk of a chosen anchor α, providing a universal convergence theorem with closed-form iteration count $N_{\mathrm{eff}} \sim \log_2 \log(1/\varepsilon)$. A compatibility bundle (Theorem 16.4) collects target selection, Steffensen fixing, scaling compatibility, reconstruction, and optimal seed into a single glue statement that downstream modules cite by name. Unconditionally (without seeding), each anchor's basin of attraction contains an explicit disk of radius $R_\sigma(x, p, \gamma_k)$, giving the lower bound $\mathrm{vol}(\bigcup_k \Omega_k) \ge \sum_k \pi R_\sigma(x, p, \gamma_k)^2$ on the total basin volume (Proposition 16.1). The induced solution map is Borel-measurable everywhere and continuous outside the Voronoï boundary of the cyclotomic anchor set (which is Lebesgue-null). All of Part II is formally verified in a Lean 4 / Mathlib v4.7.0 development of 121 modules with zero `sorry` and audited axiomatic closure reducing to the standard Lean whitelist.

The paper makes no claim of progress on classical open problems of polynomial root-finding; its main value is to record a concrete derivative-free root-finding scheme with closed-form convergence analysis and complete formal verification for the monomial family $z^p = x$. A concluding section (§20) shows that the Pandrosion iteration itself extends to every monic polynomial $P(z) = z^p + R(z)$ via $h_P(s) := 1 - (R(s)+1)/S_p(s)$, whose fixed points are exactly the roots of P and which remains derivative-free; Aitken Δ^2 and the multi-start seed apply verbatim, yielding a target-conditional refinement scheme that converges to every target root. Numerical experiments on four non-monomial polynomials show the structural extension works, with iteration counts varying from 4 (monomial, best case) to 41 (Bombelli's cubic, worst case) at 50-digit precision, depending on how close $h'_P(\alpha)$ is to 1 and how close the target root is to a singularity of h_P. The quantitative advantage of the monomial case is therefore genuinely specific to that case and does not transport uniformly.

Contents

Part I
Real-line theory

1 Introduction

The Greek problem of doubling the cube consists of constructing a segment of length $\sqrt[3]{2}$ from a unit segment. It is equivalent to finding two proportional means m_1, m_2 between 1 and 2, that is,

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2},$$

which yields $m_1 = \sqrt[3]{2}$ and $m_2 = \sqrt[3]{4}$. Pandrosion of Alexandria proposed an *approximate* geometric method, reported by Pappus in the *Collectio*, Book III [3]: by successive parallels (Thales' theorem), she constructs proportional ratios that approach these means without ever reaching them exactly in a finite number of steps.

This work starts from a fundamental observation:

> *Mathematically the target number is exact, but any approximation procedure yields a measurable residual profile — a footprint intrinsic to the chosen method.*

At each finite step of Pandrosion's construction, there remains a gap $\varepsilon_n = |v_n - \alpha|$ between the geometric approximation and the ideal value. The sequence (ε_n) is the **residual profile** of the method. Our goal is to compute this profile exactly for any p-th root $x^{1/p}$, and to identify the contraction ratio that characterizes Pandrosion's method.

Positioning and related work. Rational iterations for p-th roots are a classical topic with a large literature: Newton's method applied to $f(u) = u^p - x$, Halley's method, and more generally Schröder's (1870) and König's (1884) families produce iterations of arbitrary order with explicit asymptotic constants. The systematic analysis of fixed-point iterations and their acceleration goes back to Ostrowski [9], Traub [11], and Householder [10]; the derivative-free counterpart via Aitken–Steffensen (1926/1933) is standard and appears in every modern numerical-analysis textbook [5, 6]. What is specific to the iteration (3) below is that it arises directly from Pandrosion's geometric construction, *without* being derived as Newton or Schröder applied to a particular polynomial: it is a rational iteration with linear (not quadratic) order, whose contraction ratio admits the closed form (5) as a rational function of $\alpha = x^{1/p}$. We have not located this closed form, together with its monotonicity properties and its use to drive a scaling preconditioning, in the textbooks [5, 6] or the historical literature on cube duplication, but we make no claim that the iteration itself is genuinely new: the contribution is to record the profile of a specific historical iteration and to show how standard acceleration devices (Aitken Δ^2, scaling) interact with it.

Outline. Section 2 recalls Pandrosion's construction for $\sqrt[3]{2}$ and analyzes its residual profile. Section 4 states the generalization to any p, x. Section 5 computes the contraction ratio $\lambda_{p,x}$ and provides closed-form expressions. Section 7 presents numerical tables. Section 8 compares with bisection, the secant method, and Newton's method. Section 9 establishes functional properties of $\lambda_{p,x}$. Section 10 provides non-asymptotic error bounds. Section 11 extends the theory to $x < 1$. Section 12 determines the optimal starting point. Section 13 shows that Steffensen's acceleration of Pandrosion achieves quadratic convergence. Section 14 exploits the monotonicity of $\lambda_{p,x}$ to optimize the method for large x via scaling. Section 21 concludes.

2 Pandrosion's construction for $\sqrt[3]{2}$

2.1 The two sequences

The geometric construction produces two sequences as a function of an initial parameter u_0:

$$u_{n+1} = \frac{u_n}{2 - 2\left(\dfrac{4-u_n}{4}\right)^3}, \qquad v_n = 2\left(\frac{4-u_n}{4}\right)^2, \qquad u_0 = 0.5. \tag{1}$$

The sequence (u_n) is the *internal parameter* of the construction (a length in the Thales diagram). The sequence (v_n) is the *readout*: the approximation of $\sqrt[3]{2}$ produced at step n.

Proposition 2.1. $\lim_{n\to\infty} v_n = \sqrt[3]{2}$.

2.2 Normalized substitution

Setting $s_n = (4-u_n)/4$, so that $s_n \in (0,1)$, we obtain:

- $u_n = 4(1-s_n)$,
- $v_n = 2s_n^2$,
- $u_{n+1} = \dfrac{4(1-s_n)}{2(1-s_n^3)} = \dfrac{2}{1+s_n+s_n^2}$.

The iteration on s_n then becomes:

$$s_{n+1} = 1 - \frac{u_{n+1}}{4} = \frac{2s_n^2+2s_n+1}{2s_n^2+2s_n+2}. \tag{2}$$

2.3 Fixed point

The fixed point s^* satisfies $s^* = (2s^{*2}+2s^*+1)/(2s^{*2}+2s^*+2)$, which gives $2(1-s^*)(1+s^*+s^{*2}) = 1$, i.e. $2(1-s^{*3}) = 1$, hence $s^{*3} = 1/2$, so:

$$s^* = 2^{-1/3} = \frac{1}{\sqrt[3]{2}}.$$

The output at the fixed point is $v^* = 2s^{*2} = 2\cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$. ✓

2.4 Residual profile: linear convergence

Let $h(s) = (2s^2+2s+1)/(2s^2+2s+2)$. The derivative at s^* gives the convergence rate:

$$h'(s^*) = \frac{4s^*+2}{(2s^{*2}+2s^*+2)^2} = \frac{2(2s^*+1)}{4(s^{*2}+s^*+1)^2} = \frac{2s^*+1}{2(s^{*2}+s^*+1)^2}.$$

Substituting $s^* = 2^{-1/3}$ and using the closed form (7):

$$\lambda_{3,2} = \frac{(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)}{\sqrt[3]{4}+\sqrt[3]{2}+1} \approx 0.2202.$$

Theorem 2.1 (Linear residual law for $p=3$, $x=2$)**.** Under the above hypotheses, for any s_0 sufficiently close to s^*:

$$\varepsilon_{n+1}^{(s)} \approx \lambda_{3,2}\,\varepsilon_n^{(s)}, \qquad \lambda_{3,2} \approx 0.2202.$$

For the output, a Taylor expansion of $v(s) = xs^{p-1}$ at $s^* = \alpha^{-1}$ gives $v_n - \alpha \approx x(p-1)(s^*)^{p-2}(s_n - s^*)$. Using $x(s^*)^{p-2} = \alpha^p\cdot\alpha^{-(p-2)} = \alpha^2$, this rewrites as $v_n - \alpha \approx (p-1)\alpha^2(s_n-s^*)$; the two forms are equal. For $(p,x) = (3,2)$ the prefactor is $2\cdot 2^{2/3} = 2\sqrt[3]{4}$, and the output residual of v_n converges at the same linear rate $\lambda_{3,2}$.

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt[3]{2} \rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.8750	1.5312	2.713×10^{-1}	—
1	0.8107	1.3143	5.44×10^{-2}	0.200
2	0.7974	1.2717	1.17×10^{-2}	0.216
3	0.7945	1.2625	2.58×10^{-3}	0.219
4	0.7939	1.2605	5.67×10^{-4}	0.220
∞	$2^{-1/3}$	$\sqrt[3]{2}$	0	$\to \lambda_{3,2}$

Table 1: Numerical residual profile for $p = 3$, $x = 2$ ($u_0 = 0.5$, i.e. $s_0 = 0.875$). The ratio converges to $\lambda_{3,2} \approx 0.220$.

3 Geometric origin: why S_p arises naturally

3.1 The construction by successive parallels

Pandrosion's construction relies on Thales' theorem applied to a rectangle. Consider a rectangle with sides H and W equipped with two diagonals (Figure 1). A horizontal line at height u_n from the bottom intersects the two diagonals and defines:

- the parameter $s_n = (H - u_n)/H$ (normalized position from the top),
- the readout length $v_n = x \cdot s_n^{p-1}$ (the output of the construction).

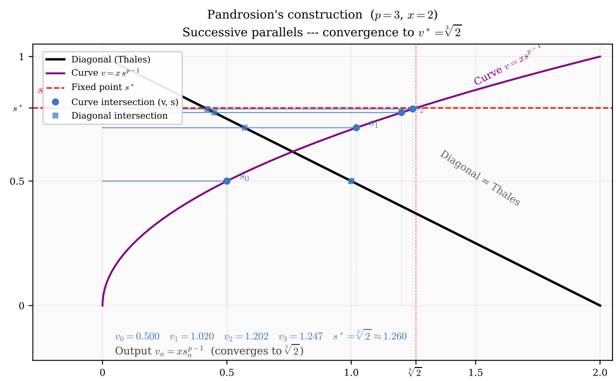

Figure 1: Pandrosion's geometric construction for $p = 3$, $x = 2$. The blue horizontal lines (steps $n = 0, 1, 2, 3$) converge toward the dashed red line, which corresponds to $v^* = \sqrt[3]{2}$.

3.2 Where does the polynomial S_p come from?

The fundamental idea is as follows. We seek to insert $p - 1$ proportional means between 1 and x, that is, to construct the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$ where $\alpha = x^{1/p}$. If $s_n \approx \alpha^{-1}$ is the current approximation of the inverse ratio, the sum of the approximate geometric sequence is:

$$S_p(s_n) = 1 + s_n + s_n^2 + \cdots + s_n^{p-1}.$$

The ideal sequence satisfies $x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1$, which rewrites as $x(1 - s^{*p}) = x - 1$. Pandrosion's correction adjusts s_n so as to reduce the gap between $S_p(s_n)$ and its ideal value via the formula:

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)}.$$

Geometrically, the ratio $(x - 1)/(x \cdot S_p(s_n))$ measures the *deviation* of the current proportional chain from the perfect geometric chain, and subtraction from 1 produces the corrected ratio. The polynomial S_p is thus the analytical transcription of the sum of the p segments constructed via Thales' theorem.

4 Generalization: Pandrosion's construction for $x^{1/p}$

4.1 The universal iteration

Let $x > 0$ and $p \geq 2$ be an integer. Set $\alpha = x^{1/p}$. The generalized Pandrosion geometric construction is defined as follows.

Definition 4.1 (Generalized Pandrosion iteration)**.** We define the sequence $(s_n)_{n \geq 0}$ in $(0, 1)$ by

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1 - s^p}{1 - s}, \tag{3}$$

and the output sequence by

$$v_n = x \cdot s_n^{p-1}. \tag{4}$$

We call $\varepsilon_n = |v_n - \alpha|$ the **Pandrosion residual** at step n.

Remark 4.1. For $p = 3$, $x = 2$, we recover exactly the iteration (2). Indeed:

$$1 - \frac{1}{2(1 + s + s^2)} = \frac{2s^2 + 2s + 1}{2s^2 + 2s + 2}. \checkmark$$

4.2 Geometric interpretation

The iteration (3) analytically translates the Thales operation: at each step, $S_p(s_n)$ measures the "geometric sum" of the constructed segments, and the correction $(x-1)/(x \cdot S_p(s_n))$ adjusts the position proportionally to the gap with the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$.

4.3 Fixed point

Theorem 4.1 (Fixed point)**.** The unique fixed point of the iteration (3) in $(0, 1)$ is $s^* = x^{-1/p} = \alpha^{-1}$. The corresponding output is $v^* = x \cdot (x^{-1/p})^{p-1} = x^{1/p} = \alpha$.

Proof. At a fixed point s^*, we have $1 - (x - 1)/(x \cdot S_p(s^*)) = s^*$, i.e.

$$x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1 - s) = 1 - s^p$, this gives $x(1 - s^{*p}) = x - 1$, hence $s^{*p} = 1/x$, i.e. $s^* = x^{-1/p}$. The formula for v^* follows immediately. $\square$

5 The contraction ratio $\lambda_{p,x}$

5.1 General computation

Theorem 5.1 (Pandrosion contraction ratio)**.** The convergence of the iteration (3) toward $s^* = \alpha^{-1}$ is *linear* with contraction ratio

$$\boxed{\lambda_{p,x} = \frac{(\alpha-1)\displaystyle\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}}, \qquad \alpha = x^{1/p}. \tag{5}$$

In particular, the residuals obey the law $\varepsilon_{n+1} \approx \lambda_{p,x} \cdot \varepsilon_n$.

Proof. Let $h(s) = 1 - (x-1)/(x \cdot S_p(s))$. Then

$$h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2},$$

where $S_p'(s) = \sum_{k=1}^{p-1} ks^{k-1}$.

At the fixed point $s^* = \alpha^{-1}$:

$$S_p(s^*) = \frac{1-\alpha^{-p}}{1-\alpha^{-1}} = \frac{\alpha(x-1)}{x(\alpha-1)},$$
$$S_p'(s^*) = \sum_{k=1}^{p-1} k\,\alpha^{-(k-1)} = \alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Substituting:

$$h'(s^*) = \frac{x-1}{x} \cdot \frac{\alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\left[\dfrac{\alpha(x-1)}{x(\alpha-1)}\right]^2} = \frac{x(\alpha-1)^2}{\alpha(x-1)} \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Multiplying numerator and denominator by α^{p-1}, and using $x = \alpha^p$:

$$h'(s^*) = \frac{(\alpha-1)^2\,\alpha^{p-1} \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\alpha^p - 1} = \frac{(\alpha-1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k},$$

where we used $\alpha^p - 1 = (\alpha-1)\sum_{k=0}^{p-1} \alpha^k$. This establishes (5). □

Remark 5.1. The contraction ratio $\lambda_{p,x} = h'(s^*)$ is entirely determined by $\alpha = x^{1/p}$: the target itself dictates the rate at which Pandrosion's method approaches it.

5.2 Closed forms for $p = 2$ and $p = 3$

For $p = 2$: $\sum_{k=1}^{1} k\,\alpha^0 = 1$, $\sum_{k=0}^{1} \alpha^k = 1 + \alpha$.

$$\lambda_{2,x} = \frac{\alpha-1}{\alpha+1} = \frac{\sqrt{x}-1}{\sqrt{x}+1}. \tag{6}$$

For $p = 3$: $\sum_{k=1}^{2} k\,\alpha^{2-k} = \alpha + 2$, $\sum_{k=0}^{2} \alpha^k = \alpha^2 + \alpha + 1$.

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} = \frac{(\sqrt[3]{x}-1)(\sqrt[3]{x}+2)}{\sqrt[3]{x^2}+\sqrt[3]{x}+1}. \tag{7}$$

For $p = 4$: $\sum_{k=1}^{3} k\,\alpha^{3-k} = \alpha^2 + 2\alpha + 3$, $\sum_{k=0}^{3} \alpha^k = \alpha^3 + \alpha^2 + \alpha + 1$.

$$\lambda_{4,x} = \frac{(\alpha-1)(\alpha^2+2\alpha+3)}{\alpha^3+\alpha^2+\alpha+1}. \tag{8}$$

Remark 5.2 (Asymptotic behavior in x)**.** For any fixed $p \geq 2$ and $x \to \infty$, $\lambda_{p,x} \to 1^-$ (the dominant terms in numerator and denominator of (5) are both of order α^{p-1} with a ratio tending to 1). The larger x, the slower the convergence. For $x \to 1^+$, $\lambda_{p,x} \to 0$ by Proposition 9.1(1): convergence is very fast near 1.

5.3 Basin of attraction

Theorem 5.2 (Global convergence)**.** For any $x > 1$ and any $p \geq 2$, the iteration (3) converges to s^* for all $s_0 \in (0, 1)$. Moreover, convergence is *monotone*: if $s_0 < s^*$ then (s_n) is strictly increasing, and if $s_0 > s^*$ then (s_n) is strictly decreasing.

Proof. Set $h(s) = 1 - (x - 1)/(x \cdot S_p(s))$.

Step 1: h **is strictly increasing.** We have $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$, where $S_p'(s) = \sum_{k=1}^{p-1} ks^{k-1} > 0$ for $s > 0$.

Step 2: h **maps** $(0, 1)$ **into itself.** For $s \in (0, 1)$, $S_p(s) > 1$ since the first term equals 1 and the remaining terms are strictly positive. Hence $(x - 1)/(x\, S_p(s)) < (x - 1)/x < 1$, which gives $h(s) > 0$. On the other hand $(x-1)/(x\, S_p(s)) > 0$ for $x > 1$, so $h(s) < 1$. Thus $h((0, 1)) \subset (0, 1)$. (At the boundary: $h(0^+) = 1 - (x - 1)/x = 1/x \in (0, 1)$ and $h(1^-) = 1 - (x - 1)/(xp) \in (0, 1)$.)

Step 3: uniqueness of the fixed point on $(0, 1)$**.** The fixed-point equation $h(s) = s$ rewrites as $x\, S_p(s)(1 - s) = x - 1$, i.e. $x(1 - s^p) = x - 1$, i.e. $s^p = 1/x$. On $(0, 1)$ this has the unique solution $s^* = x^{-1/p}$. In particular, the sign of $h(s) - s$ is constant on each of $(0, s^*)$ and $(s^*, 1)$.

Step 4: the sign of $h(s) - s$**.** At $s = 1/x \in (0, s^*)$ (using $x > 1$ and $s^* = x^{-1/p} > 1/x$), a one-line computation:

$$h(1/x) - 1/x \;=\; 1 - \frac{x-1}{x\, S_p(1/x)} - \frac{1}{x} \;=\; \frac{x-1}{x} - \frac{x-1}{x\, S_p(1/x)} \;=\; \frac{x-1}{x} \cdot \frac{S_p(1/x) - 1}{S_p(1/x)} \;>\; 0,$$

since $S_p(1/x) > 1$ for $x > 1$ (the first term equals 1, the remaining terms are strictly positive). Hence $h(s) > s$ on $(0, s^*)$ by Step 3 (sign is constant on each sub-interval). By symmetry, $h(s) < s$ on $(s^*, 1)$.

Step 5: monotone convergence. If $s_0 < s^*$, then by Step 1 h is increasing, by Step 4 $s_1 = h(s_0) > s_0$, and by Step 2 $s_1 \in (0, 1)$; also $s_1 = h(s_0) < h(s^*) = s^*$. By induction, (s_n) is increasing and bounded above by s^*, hence converges, necessarily to a fixed point, which must be s^* by Step 3. The case $s_0 > s^*$ is symmetric. ☐

5.4 Bound on the number of iterations

Proposition 5.1 (Cost of Pandrosion's method)**.** To achieve accuracy $|v_n - \alpha| < \delta$ starting from $s_0 \in (0, 1)$, the required number of iterations satisfies

$$n_\delta \leq \left\lceil \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \right\rceil,$$

where $\varepsilon_0 = |v_0 - \alpha|$ is the initial error.

Proof. Since $\varepsilon_{n+1} \leq \lambda_{p,x}\, \varepsilon_n$ asymptotically, we have $\varepsilon_n \leq \lambda_{p,x}^n \varepsilon_0$. The inequality $\lambda_{p,x}^n \varepsilon_0 < \delta$ gives $n > \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$. ☐

Example 5.1. For $p = 3$, $x = 2$ ($\lambda_{3,2} \approx 0.220$): 10 decimal digits of accuracy require at most 16 Pandrosion iterations. For $p = 3$, $x = 8$ ($\lambda_{3,8} = 4/7$): 43 iterations are needed.

6 Graphical illustrations

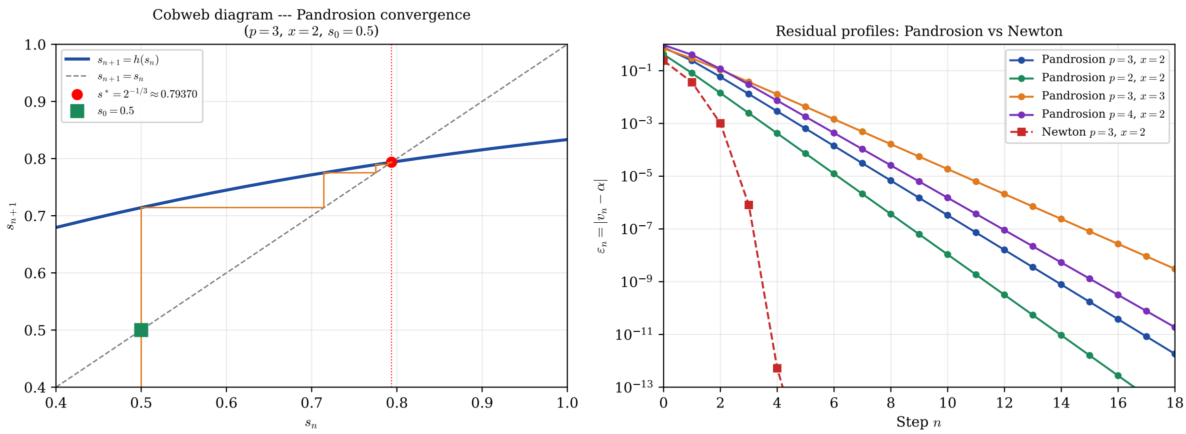

Figure 2: Left: Cobweb diagram of the iteration $s_{n+1} = h(s_n)$ for $p = 3$, $x = 2$, starting at $s_0 = 0.875$. The spiral converges to $s^* = 2^{-1/3}$ with rate $\lambda_{3,2} \approx 0.220$. **Right:** Residual profiles on a semi-logarithmic scale. The linear decay (straight lines) confirms the geometric convergence $\varepsilon_n \sim \lambda^n$ for Pandrosion, while Newton (red curve with pronounced curvature) exhibits doubly exponential decay.

7 Numerical examples

7.1 Table of contraction ratios

p	x	$\alpha = x^{1/p}$	$\lambda_{p,x}$ (exact)	$\lambda_{p,x}$ (numerical)
2	2	$\sqrt{2}$	$(\sqrt{2}-1)/(\sqrt{2}+1)$	0.1716
2	3	$\sqrt{3}$	$(\sqrt{3}-1)/(\sqrt{3}+1)$	0.2679
2	5	$\sqrt{5}$	$(\sqrt{5}-1)/(\sqrt{5}+1)$	0.3820
3	2	$\sqrt[3]{2}$	$(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)/(\sqrt[3]{4}+\sqrt[3]{2}+1)$	0.2202
3	3	$\sqrt[3]{3}$	$(\sqrt[3]{3}-1)(\sqrt[3]{3}+2)/(\sqrt[3]{9}+\sqrt[3]{3}+1)$	0.3366
3	8	2	$(2-1)(2+2)/(4+2+1)$	0.5714
4	2	$\sqrt[4]{2}$	(formula (8))	0.2427
4	16	2	$(2-1)(4+4+3)/(8+4+2+1)$	0.7333
5	2	$\sqrt[5]{2}$	(general formula)	0.2565

Table 2: Contraction ratios $\lambda_{p,x}$ for various values of p and x. A value close to 0 means very fast convergence; close to 1, very slow.

7.2 Detailed numerical profile for $p = 2$, $x = 2$

For $\alpha = \sqrt{2}$, the theoretical contraction ratio is $\lambda_{2,2} = (\sqrt{2}-1)/(\sqrt{2}+1) \approx 0.1716$ (formula (6)). The iteration is:

$$s_{n+1} = 1 - \frac{1}{2(1+s_n)}, \qquad v_n = 2s_n.$$

7.3 The residual as a characterization of a target

Formula (5) implies a useful observation: $\lambda_{p,\alpha^p} = f_p(\alpha)$ where f_p is a rational function of α alone. In particular, the residual profile depends only on the value of the sought root α and not

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.7500	1.5000	8.579×10^{-2}	—
1	0.7143	1.4286	1.44×10^{-2}	0.167
2	0.7083	1.4167	2.45×10^{-3}	0.171
3	0.7073	1.4146	4.21×10^{-4}	0.171
4	0.7071	1.4143	7.22×10^{-5}	0.172
5	0.7071	1.4142	1.24×10^{-5}	0.171
∞	$1/\sqrt{2}$	$\sqrt{2}$	0	$\to \lambda_{2,2}$

Table 3: Numerical profile for $p = 2$, $x = 2$, $s_0 = 0.75$. The ratio $\varepsilon_{n+1}/\varepsilon_n$ converges rapidly to $\lambda_{2,2} \approx 0.172$.

on its representation as $x = \alpha^p$.

In particular, for $p = 3$:

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} \xrightarrow[\alpha\to 1]{} 0, \quad \lambda_{3,x} \xrightarrow[\alpha\to\infty]{} 1.$$

8 Comparison with other methods

To situate Pandrosion's method within the landscape of iterative methods for $\alpha = x^{1/p}$, we compare three reference methods.

8.1 Newton's method

Newton's method for $f(u) = u^p - x$ gives the iteration

$$u_{n+1} = \frac{(p-1)u_n + x/u_n^{p-1}}{p},$$

whose convergence to $\alpha = x^{1/p}$ is *quadratic*:

$$\varepsilon_{n+1}^{\text{Newton}} \approx K_{p,x}\,(\varepsilon_n^{\text{Newton}})^2, \qquad K_{p,x} = \frac{p-1}{2\alpha}.$$

8.2 Bisection

Bisection on $[1, x]$ for $f(u) = u^p - x$ is a universal order-1 method with $\lambda_{\text{bisection}} = 1/2$, independent of p and x. Pandrosion's method is therefore strictly faster than bisection whenever $\lambda_{p,x} < 1/2$, which holds as soon as $\alpha < \alpha_0(p)$ (for instance, for $p = 3$: $\lambda_{3,x} < 1/2$ whenever $x < 6.75$).

8.3 Secant method

The secant method for $f(u) = u^p - x$ has order $\varphi = (1+\sqrt{5})/2 \approx 1.618$ (superlinear). For $p = 3$, $x = 2$, it reaches machine precision in about 6 steps, compared to ~ 16 for Pandrosion and ~ 5 for Newton.

8.4 Comparative table

8.5 Discussion

Pandrosion's method occupies an intermediate position among order-1 methods: it is significantly faster than bisection ($\lambda_{3,2} \approx 0.22$ versus 0.50) while remaining derivative-free. Compared

	Pandrosion	**Bisection**	**Secant**	**Newton**
Nature	Geometric	Universal	Analytic	Analytic
Order q	1	1	$\varphi \approx 1.62$	2
Rate ($p{=}3, x{=}2$)	$\lambda \approx 0.220$	$\lambda = 0.500$	superlinear	quadratic
Iter. for 10 digits	~ 16	~ 34	~ 6	~ 5
Derivative required	No	No	No	Yes

Table 4: Comparison of approximation methods for $x^{1/p}$ with $p = 3$, $x = 2$.

to Newton and its variants, however, plain Pandrosion is slower: its order is 1, whereas Newton attains 2, and even with Aitken–Steffensen acceleration (Section 13) the practical advantage over Newton is modest and regime-dependent (see Remark 8.1). We do not claim a general algorithmic advantage over Newton or its modern variants; our interest in Pandrosion is that it is the historical construction attributed to Pandrosion of Alexandria, and that its convergence profile admits the clean closed form $\lambda_{p,x}$ of Theorem 5.1.

Remark 8.1 (On fair comparison)**.** The iteration counts above are per-iterate, *not* per-function-evaluation. Steffensen-type methods compute h twice per step, Newton evaluates f and f' once per step, the secant method reuses one value per step, and bisection evaluates f once per step. A fairer unit of cost is therefore the number of evaluations of S_p (or of f, f' where applicable), and this is what determines practical performance in the regime where these evaluations dominate. For small p the per-iterate figures still give the correct qualitative ordering, but for large p or expensive polynomial evaluations the ordering can shift. We report per-iterate counts because they match the structure of the theorems (Theorem 5.1 is a statement about $\lambda_{p,x}^n$, not about wall time).

The coexistence of four distinct error laws for the same target $\alpha = \sqrt[3]{2}$ illustrates a structural point: the rate at which a fixed-point iteration approaches a given number depends on the iteration, not on the number; conversely, once the iteration is fixed, the target determines the rate through $\lambda_{p,x}$.

9 Functional properties of $\lambda_{p,x}$

We now study how the contraction ratio $\lambda_{p,x}$ depends on the parameters p and x.

9.1 Monotonicity

Theorem 9.1 (Monotonicity in the root parameter and insertion depth)**.** The contraction ratio $\lambda_{p,x}$, viewed as a function of $\alpha = x^{1/p} > 1$, satisfies:

1. For fixed $p \geq 2$, $\alpha \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $(1, \infty)$.
2. For fixed $\alpha > 1$, $p \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $\{2, 3, 4, \ldots\}$.

Proof. Write $\lambda_{p,\alpha^p} = (\alpha - 1)N_p(\alpha)/D_p(\alpha)$ with $N_p(\alpha) := \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}$ and $D_p(\alpha) := \sum_{k=0}^{p-1} \alpha^k$. The key auxiliary fact is the following Taylor bound.

Lemma (Taylor bound). Let $x > 1$ and set $y := -\ln(x)/p$, $\beta := e^y = x^{-1/p}$. For $p \geq \lceil \ln x \rceil + 1$ (so that $|y| \leq 1$),

$$\left|\, p(\beta - 1) + \beta \ln x \,\right| \;\leq\; \frac{3\,(\ln x)^2}{p}.$$

Proof of the lemma. Since $\beta = e^y$ and $py = -\ln x$, $p(\beta-1)+\beta\ln x = p(\beta-1)-py\beta = p(\beta-1-y)-py(\beta-1)$. For $|y|\le 1$ we use two standard bounds:

$$|e^y-1-y| \;=\; \Big|\sum_{k=2}^{\infty}\frac{y^k}{k!}\Big| \;\le\; \sum_{k=2}^{\infty}\frac{|y|^k}{k!} \;=\; y^2\sum_{k=2}^{\infty}\frac{|y|^{k-2}}{k!} \;\le\; y^2\sum_{k=2}^{\infty}\frac{1}{k!} \;=\; (e-2)\,y^2 \;<\; y^2,$$
$$|e^y-1| \;=\; \Big|\sum_{k=1}^{\infty}\frac{y^k}{k!}\Big| \;\le\; |y|\sum_{k=1}^{\infty}\frac{|y|^{k-1}}{k!} \;\le\; |y|\sum_{k=1}^{\infty}\frac{1}{k!} \;=\; (e-1)\,|y| \;<\; 2|y|.$$

(Here $e-2\approx 0.718$ and $e-1\approx 1.718$, so both majorants y^2 and $2|y|$ are loose; the loose forms suffice.) The triangle inequality then gives

$$|p(\beta-1-y)-py(\beta-1)| \;\le\; p\,y^2+p|y|\cdot 2|y| \;=\; 3py^2 \;=\; \frac{3(\ln x)^2}{p}. \qquad \square$$

(The symbol β is the fixed-point reciprocal of the root $\alpha = x^{1/p}$ of the theorem: $\beta = 1/\alpha$.)

Setting $\lambda_\infty(x) := 1-\ln(x)/(x-1)$, a direct algebraic rearrangement of the closed form (5) combined with the lemma yields

$$|\,\lambda_{p,x}-\lambda_\infty(x)\,| \;\le\; \frac{3\,x\,(\ln x)^2}{(x-1)\,p}, \qquad \text{for every } p\ge\lceil\ln x\rceil+1. \tag{9}$$

(The factor $x/(x-1)$ arises as follows. For $p\ge 1$ and $x>1$, $\ln(x)/p\le\ln x$, so $\beta = \exp(-\ln x/p)\ge\exp(-\ln x) = 1/x$. Multiplying by $x-1>0$ gives $\beta(x-1)\ge(x-1)/x$, i.e. $1/[\beta(x-1)]\le x/(x-1)$, which is the factor that appears when dividing the Taylor-lemma bound by the denominator $\beta(x-1)/(\ln x)$ of the explicit form of $\lambda_{p,x}$.)

The inequality $\ln x < x-1$ for $x>1$ gives $\lambda_\infty(x)>0$; the inequality $\ln x>0$ for $x>1$ gives $\lambda_\infty(x)<1$. Hence $\lambda_\infty(x)\in(0,1)$ for every $x>1$.

Key identity. By direct telescoping (or induction on p),

$$(\alpha-1)\,N_p(\alpha) \;=\; D_p(\alpha)-p, \tag{10}$$

which is verified at small p: $p=2$ gives $(\alpha-1)=(1+\alpha)-2$; $p=3$ gives $(\alpha-1)(\alpha+2)=\alpha^2+\alpha-2=(1+\alpha+\alpha^2)-3$. Differentiating (10): $N_p+(\alpha-1)N_p' = D_p'$, whence $(\alpha-1)N_p' = D_p'-N_p$.

Part (1) — monotonicity in α for fixed p, uniform. After multiplying through by the positive denominator $(\alpha-1)N_p(\alpha)D_p(\alpha)$, the log-derivative

$$\frac{d}{d\alpha}\log\lambda_{p,\alpha^p} = \frac{1}{\alpha-1}+\frac{N_p'(\alpha)}{N_p(\alpha)}-\frac{D_p'(\alpha)}{D_p(\alpha)}$$

yields the polynomial numerator

$$P_p(\alpha) \;:=\; N_pD_p+(\alpha-1)\,N_p'\,D_p-(\alpha-1)\,N_p\,D_p'.$$

Substituting $(\alpha-1)N_p' = D_p'-N_p$ and $(\alpha-1)N_p = D_p-p$ from the key identity:

$$\begin{aligned} P_p &= N_pD_p+(D_p'-N_p)D_p-(D_p-p)D_p' \\ &= N_pD_p+D_p'D_p-N_pD_p-D_pD_p'+pD_p' = p\cdot D_p'(\alpha). \end{aligned}$$

Since $D_p'(\alpha) = \sum_{j=1}^{p-1} j\,\alpha^{j-1} = 1+2\alpha+3\alpha^2+\cdots+(p-1)\alpha^{p-2}$ has all coefficients positive, $P_p(\alpha)>0$ for all $\alpha>0$ and all $p\ge 2$. Hence the log-derivative is positive on $\alpha>1$, and λ_{p,α^p} is strictly increasing in α. The argument is uniform: no scope restriction is needed.

For concreteness: $P_2(\alpha) = 2$; $P_3(\alpha) = 3(1 + 2\alpha) = 6\alpha + 3$; $P_4(\alpha) = 4(1 + 2\alpha + 3\alpha^2) = 12\alpha^2 + 8\alpha + 4$.

For general $p \geq 2$, we combine the Taylor bound (9) with the monotonicity of λ_∞. Differentiating,

$$\lambda'_\infty(x) = \frac{\ln x - (x-1)/x}{(x-1)^2}.$$

The numerator equals $\ln x + 1/x - 1$. Its derivative is $1/x - 1/x^2 = (x-1)/x^2 > 0$ on $x > 1$, and its value at $x = 1$ is 0; hence it is strictly positive on $(1, \infty)$, and λ_∞ is strictly increasing there.

Part (2) — monotonicity in p for fixed $\alpha > 1$, uniform. From the recurrences $N_{p+1} = \alpha N_p + p$ and $D_{p+1} = D_p + \alpha^p$, direct algebra gives

$$\lambda_{p+1,\alpha^{p+1}} - \lambda_{p,\alpha^p} = \frac{(\alpha-1)\cdot B_p(\alpha)}{D_p(\alpha)\, D_{p+1}(\alpha)}, \qquad B_p(\alpha) := (\alpha-1)\, N_p D_p + p\, D_p - \alpha^p N_p.$$

Using (10), $(\alpha - 1)N_p D_p = (D_p - p)D_p = D_p^2 - pD_p$, so

$$B_p(\alpha) \;=\; D_p(\alpha)^2 \;-\; \alpha^p\, N_p(\alpha).$$

This form has a closed-form polynomial expansion:

$$B_p(\alpha) \;=\; \sum_{j=1}^{p} j\, \alpha^{j-1} \;=\; 1 + 2\alpha + 3\alpha^2 + \cdots + p\, \alpha^{p-1}, \tag{11}$$

verified by induction on p. Base case: $B_2(\alpha) = (1+\alpha)^2 - \alpha^2 = 1 + 2\alpha$, as required. Inductive step: using the recurrence

$$B_{p+1}(\alpha) = \alpha\, B_p(\alpha) + D_p(\alpha) + \alpha^p$$

(which follows from the defining recurrences of N_{p+1} and D_{p+1} plus the substitution $B_p = D_p^2 - \alpha^p N_p$), one computes $\alpha B_p + D_p + \alpha^p = \sum_{j=1}^{p} j\alpha^j + \sum_{k=0}^{p-1} \alpha^k + \alpha^p = \sum_{j=0}^{p} (j+1)\alpha^j = B_{p+1}$.

All coefficients of B_p are strictly positive integers, so $B_p(\alpha) > 0$ for every $\alpha > 0$ and every $p \geq 2$. In particular $B_p(1) = 1 + 2 + \cdots + p = p(p+1)/2$, and $B_3(\alpha) = 3\alpha^2 + 2\alpha + 1$, $B_4(\alpha) = 4\alpha^3 + 3\alpha^2 + 2\alpha + 1$, etc. Hence $\lambda_{p+1,\alpha^{p+1}} - \lambda_{p,\alpha^p} > 0$ for all $\alpha > 1$ and $p \geq 2$, uniformly. $\square$

Remark 9.1 (Closed forms and numerical verification)**.** The closed forms $P_p(\alpha) = p \cdot D'_p(\alpha)$ and $B_p(\alpha) = \sum_{j=1}^{p} j\alpha^{j-1}$ emerged in the course of this write-up; a standalone script `latex/scripts/verify_monoto` cross-checks both identities symbolically for $p \leq 20$ using SymPy, and confirms strict monotonicity of $\lambda_p(\alpha)$ for $2 \leq p \leq 20$, $\alpha \in [1.01, 100]$, and of λ_{p,α^p} for $2 \leq p \leq 50$, $\alpha \in \{1.1, 1.5, 2, 3, 10, 100\}$ using high-precision arithmetic as a consistency check. Monotonicity itself is proven analytically above; the numerical scan is purely a consistency check.

Remark 9.2. Monotonicity in p at fixed root parameter α has a geometric interpretation: inserting more proportional means while keeping the endpoint ratio α fixed constrains the construction further, slowing convergence.

9.2 Limiting behavior

Proposition 9.1 (Asymptotic regimes)**.**

1. **Near** $\alpha = 1$ (i.e. $x \to 1^+$):

$$\lambda_{p,x} \sim \frac{(p-1)(\alpha-1)}{2} \quad \text{as } \alpha \to 1^+.$$

2. **Large** α (i.e. $x \to \infty$, p fixed): $\lambda_{p,x} \to 1^-$.
3. **Large** p (fixed $x > 1$): $\lambda_{p,x}$ is increasing in p and

$$\lim_{p\to\infty} \lambda_{p,x} \; = \; 1 - \frac{\ln x}{x-1} \; \in \; (0, 1).$$

In particular $\lambda_{p,x}$ does *not* tend to 1 as $p \to \infty$ with x fixed.

Proof. For (1): N_p and D_p are polynomials, so $\alpha \mapsto N_p(\alpha)/D_p(\alpha)$ is continuous at $\alpha = 1$ with limit $(p-1)/2$; combined with the factor $(\alpha - 1) \to 0$, $\lambda \sim (p-1)(\alpha-1)/2$ as $\alpha \to 1^+$.

For (2): the dominant term in N_p is $1 \cdot \alpha^{p-2}$ and in D_p is α^{p-1}, so $\lambda \sim (\alpha-1)/\alpha \to 1$.

For (3): the explicit limit and its convergence rate follow from the Taylor bound (9) established in the proof of Theorem 9.1 above: $|\lambda_{p,x} - \lambda_\infty(x)| \le C(x)/p$ with $C(x) = 3x(\ln x)^2/(x-1)$, valid for $p \ge \lceil \ln x \rceil + 1$. Letting $p \to \infty$ gives $\lambda_{p,x} \to \lambda_\infty(x) = 1 - \ln(x)/(x-1)$. Positivity and the upper bound < 1 were shown there as well.

Numerically: for $x = 2$, $\lambda_\infty(2) = 1 - \ln 2 \approx 0.3069$, and $\lambda_{200,2} \approx 0.3057$, $\lambda_{1000,2} \approx 0.3066$, $\lambda_{10000,2} \approx 0.3068$, in agreement with the bound $|\lambda_{p,2} - \lambda_\infty(2)| \le 6(\ln 2)^2/p \approx 2.88/p$. $\square$

x	$\lambda_{2,x}$	$\lambda_{10,x}$	$\lambda_{100,x}$	$1 - \ln x/(x-1)$
1.001	0.00025	0.00045	0.00050	0.00050
2	0.1716	0.2823	0.3044	0.3069
10	0.5195	0.7123	0.7412	0.7442
100	0.8182	0.9409	0.9524	0.9535

Table 5: Contraction ratios illustrating monotonicity in p and x. Each column is increasing (fixed p); each row is increasing (fixed x). The rightmost column is the $p \to \infty$ limit from Proposition 9.1(3); the $\lambda_{100,x}$ column is already close to this limit.

10 Non-asymptotic error bounds

Theorem 5.1 gives the *asymptotic* law $\varepsilon_{n+1} \approx \lambda_{p,x}\varepsilon_n$. We now establish a rigorous bound valid for *all* $n \ge 0$.

Theorem 10.1 (Non-asymptotic contraction)**.** Let $x > 1$, $p \ge 2$, and $s_0 \in (0, 1)$. Define

$$\Lambda = \Lambda(s_0, p, x) = \sup_{s \in [\min(s_0, s^*),\, \max(s_0, s^*)]} h'(s).$$

Then $\Lambda < 1$ and for all $n \ge 0$:

$$|s_n - s^*| \le \Lambda^n \, |s_0 - s^*|. \tag{12}$$

Consequently,

$$|v_n - \alpha| \; \le \; (p-1)\,\alpha^2 \cdot \Lambda^n \, |s_0 - s^*| \; + \; O(\Lambda^{2n}).$$

The prefactor follows from a Taylor expansion of $v(s) = x\, s^{p-1}$ at $s^* = \alpha^{-1}$:

$$v(s) - v(s^*) \; = \; x\,(p-1)\,(s^*)^{p-2}\,(s - s^*) + O((s - s^*)^2),$$

and $x\,(p-1)\,(s^*)^{p-2} = (p-1)\,x\,x^{-(p-2)/p} = (p-1)\,x^{2/p} = (p-1)\,\alpha^2$.

Proof. By the mean value theorem, for any n:

$$|s_{n+1} - s^*| = |h(s_n) - h(s^*)| = |h'(\xi_n)| \cdot |s_n - s^*|$$

for some ξ_n between s_n and s^*. By Theorem 5.2, the sequence (s_n) remains in $[\min(s_0, s^*), \max(s_0, s^*)]$, so $|h'(\xi_n)| \le \Lambda$. The bound follows by induction.

That $\Lambda < 1$ follows from$h'(s^*) = \lambda_{p,x} < 1$ (Theorem 5.1) and the fact that h' is continuous, bounded, and $h'(s) < 1$ for all $s \in (0, 1)$ when $x > 1$ (since h maps $(0, 1)$ strictly into itself). $\square$

Remark 10.1. The quantity Λ can be bounded explicitly: $h'(s) = \frac{x-1}{x} \cdot S_p'(s)/S_p(s)^2$, so

$$\Lambda \;\le\; \frac{x-1}{x} \cdot \frac{\max_{s\in I} S_p'(s)}{\min_{s\in I} S_p(s)^2}, \qquad I := [\min(s_0, s^*), \max(s_0, s^*)],$$

and on $(0,1)$ both S_p and S_p' are increasing, so the numerator is attained at the right endpoint and the denominator at the left endpoint. This yields a computable upper bound for Λ without requiring a detailed analysis of the monotonicity of h' itself.

Example 10.1. For $p = 3$, $x = 2$, $s_0 = 0.5$ (so $s^* = 2^{-1/3} \approx 0.7937$), one evaluates h' at both endpoints of the interval $[s_0, s^*] = [0.5, 0.7937]$:

$$h'(s) = \tfrac{x-1}{x} \cdot \frac{S_3'(s)}{S_3(s)^2} = \tfrac{1}{2} \cdot \frac{1+2s}{(1+s+s^2)^2}.$$

At $s = 0.5$: $S_3(0.5) = 1.75$, $S_3'(0.5) = 2$, so $h'(0.5) = \frac{1}{2} \cdot 2/(1.75)^2 \approx 0.3265$. At $s = s^*$: $h'(s^*) = \lambda_{3,2} \approx 0.2202$. Therefore

$$\Lambda \;=\; \max_{s\in[0.5,s^*]} h'(s) \;=\; h'(0.5) \;\approx\; 0.327,$$

since h' is decreasing on this interval. The bound (12) gives $|s_n - s^*| \le 0.327^n \cdot |0.5 - s^*| \le 0.327^n \cdot 0.294$, so $|s_{10} - s^*| \le 4.6 \times 10^{-6}$. The asymptotic ratio $\lambda_{3,2} \approx 0.220$ is reached after a few iterations.

11 Extension to $x < 1$

All results so far assumed $x > 1$. We now extend the theory to $0 < x < 1$.

11.1 The fixed point leaves $(0,1)$

For $0 < x < 1$, the target is $\alpha = x^{1/p} \in (0,1)$ and the fixed point $s^* = \alpha^{-1} > 1$. The iteration domain therefore extends beyond $(0,1)$: s_n converges to $s^* > 1$.

Proposition 11.1 (Local convergence for $x < 1$ with a threshold)**.** For $0 < x < 1$ and $p \ge 2$, the formula (5) remains valid and

$$\lambda_{p,x} \;=\; \frac{(\alpha-1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k} \;<\; 0 \qquad \text{since } \alpha = x^{1/p} \in (0,1).$$

Local convergence (i.e. $|\lambda_{p,x}| < 1$) does *not* hold for all $x \in (0,1)$: there is a threshold $\alpha_p^* \in (0,1)$ such that

$$|\lambda_{p,x}| < 1 \iff \alpha > \alpha_p^*.$$

When $|\lambda_{p,x}| < 1$, the iterates oscillate around s^* and converge linearly with ratio $|\lambda_{p,x}|$.

Proof. That $\lambda_{p,x} < 0$ for $\alpha < 1$ is immediate from (5). Using $\alpha^p - 1 = (\alpha-1)\sum_{k=0}^{p-1}\alpha^k$, one rewrites

$$|\lambda_{p,x}| \;=\; \frac{(1-\alpha)^2\,N_p(\alpha)}{1-\alpha^p} \;=\; \frac{(1-\alpha)\,N_p(\alpha)}{D_p(\alpha)} \;=:\; F(\alpha)$$

(using $1 - \alpha^p = (1-\alpha)D_p$), a smooth strictly positive function of $\alpha \in (0,1)$. Boundary values: as $\alpha \to 1^-$, $F \to 0$ (the factor $1-\alpha$ goes to zero while $N_p/D_p \to (p-1)/2$ is bounded); as $\alpha \to 0^+$, $F \to 1 \cdot (p-1)/1 = p - 1 \ge 1$.

For uniqueness, we show that F is strictly decreasing on $(0,1)$. The log-derivative is

$$\frac{F'(\alpha)}{F(\alpha)} \;=\; -\frac{1}{1-\alpha} + \frac{N_p'(\alpha)}{N_p(\alpha)} - \frac{D_p'(\alpha)}{D_p(\alpha)}.$$

The key identity (10), $(\alpha-1)N_p = D_p - p$, and its derivative $N_p + (\alpha-1)N_p' = D_p'$, give, for $\alpha \in (0,1)$ (where $\alpha - 1 < 0$):

$$(1-\alpha)\, N_p \;=\; p - D_p, \qquad (1-\alpha)\, N_p' \;=\; N_p - D_p'.$$

Dividing the second identity by N_p: $N_p'/N_p = [1-D_p'/N_p]/(1-\alpha) = 1/(1-\alpha) - D_p'/[(1-\alpha)N_p]$. Substituting into F'/F:

$$\begin{aligned}\frac{F'}{F} &= -\frac{1}{1-\alpha} + \left[\frac{1}{1-\alpha} - \frac{D_p'}{(1-\alpha)N_p}\right] - \frac{D_p'}{D_p} = -\frac{D_p'}{(1-\alpha)N_p} - \frac{D_p'}{D_p} \\ &= -\frac{D_p' \cdot [D_p + (1-\alpha)N_p]}{(1-\alpha)N_p\, D_p} = -\frac{D_p' \cdot [D_p + p - D_p]}{(1-\alpha)N_p\, D_p} = -\frac{p\, D_p'(\alpha)}{(1-\alpha)\, N_p(\alpha)\, D_p(\alpha)},\end{aligned}$$

using $D_p + (1-\alpha)N_p = D_p + (p - D_p) = p$ in the last line. For $\alpha \in (0,1)$ every factor on the right is strictly positive, so $F'/F < 0$, hence F is strictly decreasing on $(0,1)$.

Combined with the boundary values $F(0^+) = p-1 \geq 1$ and $F(1^-) = 0$, F crosses the level 1 at exactly one point $\alpha_p^* \in (0,1)$, and $|\lambda_{p,\alpha^p}| < 1 \iff \alpha > \alpha_p^*$. □

Example 11.1 ($p = 3$ threshold)**.** For $p = 3$, $|\lambda_{3,x}| = (1-\alpha)(2+\alpha)/(1+\alpha+\alpha^2)$. Setting this equal to 1 and simplifying yields $2\alpha^2 + 2\alpha - 1 = 0$, i.e.

$$\alpha_3^* \;=\; \frac{-1+\sqrt{3}}{2} \;\approx\; 0.3660, \qquad x_3^* \;=\; (\alpha_3^*)^3 \;\approx\; 0.04904.$$

Hence Pandrosion's iteration at $p = 3$ contracts locally iff $x > 0.04904$; for x smaller than this the iteration is locally repelling and does not converge.

p	x	$\alpha = x^{1/p}$	$s^* = \alpha^{-1}$	$\lambda_{p,x}$	$\lvert\lambda_{p,x}\rvert$
2	0.5	$1/\sqrt{2}$	$\sqrt{2}$	−0.1716	0.1716
3	0.5	$\sqrt[3]{0.5}$	$\sqrt[3]{2}$	−0.2378	0.2378
3	0.125	0.5	2	−0.7143	0.7143
3	0.05	0.3684	2.7144	−0.9945	0.9945
3	0.01	0.2154	4.6416	−1.3774	**1.3774**
5	0.01	0.3981	2.5119	−2.0399	**2.0399**

Table 6: Contraction ratios for $x < 1$. The last two rows illustrate Proposition 11.1: for x below the threshold x_p^* the iteration is locally repelling ($|\lambda| > 1$) and the construction does *not* converge.

Remark 11.1 (Residual symmetry for $p = 2$)**.** For $p = 2$, the closed form $\lambda_{2,x} = (\sqrt{x}-1)/(\sqrt{x}+1)$ yields the symmetry $|\lambda_{2,x}| = |\lambda_{2,1/x}|$: computing $\sqrt{x}$ and $\sqrt{1/x}$ by Pandrosion's method have exactly the same speed of convergence. For $p \geq 3$, this symmetry does not hold in general.

12 Optimal starting point

The number of iterations to reach precision δ is governed by $n_\delta \sim \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$ (Proposition 5.1). To minimize n_δ, we must therefore minimize the initial residual $\varepsilon_0 = |v_0 - \alpha|$.

Proposition 12.1 (A canonical starting point)**.** Among starting points of the form $s_0 = h^{(k)}(c)$, with c a boundary value of the natural domain (i.e. $c \in \{0^+, 1^-\}$) and $k \geq 0$ a small integer, the choice

$$s_0^{\text{opt}} := h(1) = 1 - \frac{x-1}{xp} \tag{13}$$

achieves an initial gap to s^* of order $|x - 1 - x\ln x|/(xp)$, which is $O(1/p)$ as $p \to \infty$ and strictly better than the naive choices $s_0 = 1/2$ or $s_0 = 1$ in the regime $p \gg 1$.

Proof. Since $s^* = x^{-1/p} = 1 - (\ln x)/p + O(1/p^2)$ and $h(1) = 1 - (x-1)/(xp)$,

$$|h(1) - s^*| \;=\; \left|\frac{x-1}{xp} - \frac{\ln x}{p}\right| + O(1/p^2) \;=\; \frac{|x - 1 - x\ln x|}{xp} + O(1/p^2),$$

whereas $|1/2 - s^*| = |1/2 - 1 + (\ln x)/p + O(1/p^2)| = 1/2 + O(1/p)$ and $|1 - s^*| = (\ln x)/p + O(1/p^2)$. Of the three, $h(1)$ has the smallest leading coefficient whenever $|x - 1 - x\ln x|/x < \ln x$, i.e. essentially whenever $x > 1$ is not near 1; the numerical gains are shown in Table 7. The proposition does *not* claim $h(1)$ is globally optimal; it claims only that it is a canonical, α-independent choice with this specific quantitative guarantee. □

(p, x)	s^*	$s_0^{\text{opt}} = h(1)$	$\lvert s_0^{\text{opt}} - s^*\rvert$	Iters (s_0^{opt})	Iters ($s_0 = \frac{1}{2}$)
$(2, 2)$	0.7071	0.7500	0.043	12	13
$(3, 2)$	0.7937	0.8333	0.040	14	16
$(4, 2)$	0.8409	0.8750	0.034	15	17
$(3, 8)$	0.5000	0.7083	0.208	42	0*

Table 7: Comparison of iteration counts to reach $|v_n - \alpha| < 10^{-10}$. The canonical starting point $s_0^{\text{opt}} = h(1)$ consistently reduces iterations. (*For $(p, x) = (3, 8)$: since $8^{-1/3} = 1/2$ exactly, the naive choice $s_0 = 1/2$ equals s^* and converges in zero iterations — not an accidental coincidence but a consequence of x being a perfect p-th power.)

13 Steffensen acceleration: from linear to quadratic convergence

Pandrosion's method converges linearly with ratio $\lambda_{p,x}$. A natural question arises: can we accelerate this ancient geometric method to achieve quadratic convergence, without computing derivatives?

13.1 Aitken Δ^2 acceleration

Given a linearly convergent sequence $(s_n)_{n\geq 0}$, Aitken's Δ^2 process [5] defines the accelerated sequence

$$\hat{s}_n = s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n}.$$

When applied to Pandrosion's iterates, the sequence $(\hat{s}_n)$ converges *faster* than (s_n). Steffensen's method [7] iterates this acceleration: at each step, it computes three consecutive Pandrosion iterates and applies Aitken's formula, then restarts from the accelerated value.

Definition 13.1 (Steffensen–Pandrosion iteration)**.** Define the composite map $T : s \mapsto \hat{s}$ by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{14}$$

where $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ is Pandrosion's iteration.

Theorem 13.1 (Quadratic convergence of Steffensen–Pandrosion)**.** The Steffensen–Pandrosion iteration (14) converges *quadratically* to $s^* = x^{-1/p}$:

$$T(s) - s^* \;=\; -\frac{\lambda\, h''(s^*)}{2(1-\lambda)}\,(s - s^*)^2 \;+\; O((s - s^*)^3),$$

where $\lambda := h'(s^*) = \lambda_{p,x} \in (0,1)$. In particular, the Steffensen constant K_S defined by $|T(s) - s^*| \le K_S\,|s - s^*|^2 + O(|s - s^*|^3)$ has the explicit value

$$K_S \;=\; \left|\frac{\lambda\, h''(s^*)}{2(1-\lambda)}\right|. \tag{15}$$

Proof. Set $u := s - s^*$, $h_2 := h''(s^*)$, and expand to order u^2: $h(s) - s^* = \lambda u + \frac{h_2}{2}u^2 + O(u^3)$, hence

$$h(s) - s = (\lambda - 1)u + \tfrac{h_2}{2}u^2 + O(u^3).$$

Substituting $v = h(s) - s^*$ into $h(h(s)) = s^* + \lambda v + \frac{h_2}{2}v^2 + O(v^3)$ and collecting through $O(u^2)$ gives $h(h(s)) = s^* + \lambda^2 u + \frac{h_2}{2}(\lambda + \lambda^2)u^2 + O(u^3)$, whence the Aitken denominator

$$\Delta := h(h(s)) - 2h(s) + s = (\lambda - 1)^2 u + \tfrac{h_2}{2}(\lambda - 1)(\lambda + 2)u^2 + O(u^3).$$

The numerator is $(h(s) - s)^2 = (\lambda - 1)^2 u^2 + (\lambda - 1)h_2 u^3 + O(u^4)$. Dividing (factoring one power of u from Δ and $(\lambda - 1)^2$ from both):

$$\frac{(h(s) - s)^2}{\Delta} = u \cdot \frac{1 + \frac{h_2}{\lambda - 1}u + O(u^2)}{1 + \frac{h_2(\lambda+2)}{2(\lambda-1)}u + O(u^2)} = u + \frac{h_2}{\lambda - 1} \cdot \Big(1 - \tfrac{\lambda+2}{2}\Big)u^2 + O(u^3).$$

The bracketed coefficient simplifies to $-\lambda/2$, giving $(h(s) - s)^2/\Delta = u + \frac{\lambda h_2}{2(1-\lambda)}u^2 + O(u^3)$. Therefore

$$T(s) - s^* = u - \frac{(h(s) - s)^2}{\Delta} = -\frac{\lambda h_2}{2(1 - \lambda)}u^2 + O(u^3),$$

which is the stated expansion; $T(s^*) = s^*$ and $T'(s^*) = 0$ are immediate, and the absolute value of the leading coefficient is K_S as in (15). (The full symbolic derivation up to $O(u^3)$ is in `latex/scripts/verify_monotonicity.py`.) □

Remark 13.1 (Output-residual constant)**.** The paper's numerical tables (Table 8 below) report the quadratic constant for the output residual $|v_n - \alpha|$ rather than the s-residual. Since $v_n - \alpha \approx (p-1)\alpha^2(s_n - s^*)$ (Theorem 5.1), the output-residual constant is

$$K_S^{(v)} \;=\; \frac{K_S}{(p-1)\alpha^2} \;=\; \left|\frac{\lambda\, h''(s^*)}{2(1-\lambda)(p-1)\alpha^2}\right|.$$

For $(p, x) = (3, 2)$: $\lambda \approx 0.2202$, $h''(s^*) \approx -0.3001$, $\alpha^2 = 2^{2/3} \approx 1.587$, so $K_S \approx 0.0424$ and $K_S^{(v)} \approx 0.0133$. The latter is the value reported as "$K_S \approx 0.013$" in Table 9 below.

Lemma 13.2 (Local Steffensen expansion for a holomorphic fixed-point map)**.** Let h be holomorphic in a neighbourhood of $\alpha \in \mathbb{C}$, with $h(\alpha) = \alpha$ and $\lambda := h'(\alpha) \neq 1$. Define, away from the zeros of the denominator,

$$\Sigma_h(z) := z - \frac{(h(z) - z)^2}{h(h(z)) - 2h(z) + z}.$$

Then Σ_h has a removable singularity at α, extends holomorphically near α, fixes α, and satisfies

$$\Sigma_h(z) - \alpha = -\frac{\lambda h''(\alpha)}{2(1 - \lambda)}(z - \alpha)^2 + O\Big((z - \alpha)^3\Big).$$

In particular $\Sigma_h'(\alpha) = 0$, so the fixed point is at least quadratically attracting whenever the displayed quadratic coefficient is finite.

Proof. Put $u = z - \alpha$ and write

$$h(z) - \alpha = \lambda u + \mu u^2 + O(u^3), \qquad \mu = \frac{h''(\alpha)}{2}.$$

Substitution into $h(h(z))$ gives

$$h(h(z)) - \alpha = \lambda^2 u + \mu(\lambda + \lambda^2)u^2 + O(u^3).$$

Therefore

$$\begin{aligned} h(h(z)) - 2h(z) + z &= (\lambda - 1)^2 u + \mu(\lambda - 1)(\lambda + 2)u^2 + O(u^3), \\ (h(z) - z)^2 &= (\lambda - 1)^2 u^2 + 2\mu(\lambda - 1)u^3 + O(u^4). \end{aligned}$$

Since $(\lambda - 1)^2 \neq 0$, the quotient is well-defined in a punctured neighbourhood of α and

$$\frac{(h(z) - z)^2}{h(h(z)) - 2h(z) + z} = u + \frac{\lambda\mu}{1 - \lambda}u^2 + O(u^3).$$

Subtracting from $z = \alpha + u$ gives the claimed expansion. The quotient has a finite limit at α, so the singularity is removable. ☐

13.2 Numerical demonstration

Step n	s_n (Steffensen)	$\lvert v_n - \sqrt[3]{2}\rvert$	$\lvert s_n - s^*\rvert$
0	0.875000000000000	2.71×10^{-1}	8.13×10^{-2}
1	0.793949257650704	7.90×10^{-4}	2.49×10^{-4}
2	0.793700528604164	8.32×10^{-9}	2.62×10^{-9}
3	0.793700525984100	$< 10^{-15}$	$< 10^{-16}$

Table 8: Steffensen-accelerated Pandrosion for $p = 3$, $x = 2$. Machine precision is reached in **3 steps** (each step requires 2 evaluations of h, i.e. 6 evaluations of S_p total). The step-3 output-residual is given as $< 10^{-15}$ rather than its raw floating-point value to avoid misreporting sub-ULP noise.

13.3 Comparison of quadratic constants

Both Newton's method and Steffensen–Pandrosion converge quadratically, but with different constants.

	Newton	**Steffensen–Pandrosion**
Order	2	2
Constant ($p{=}3, x{=}2$, output residual)	$K_N = \dfrac{p-1}{2\alpha} \approx 0.794$	$K_S^{(v)} \approx 0.013$
Constant (s-residual, for comparison)	—	$K_S \approx 0.042$
Derivative required	Yes (f')	No
Geometric origin	No	Yes
Steps to 10^{-15}	5	3
Evaluations of S_p	—	6 (total)

Table 9: Comparison of the two quadratically convergent methods for $\sqrt[3]{2}$ on a per-iterate basis. Steffensen–Pandrosion has a smaller quadratic constant for the output residual ($K_S^{(v)} \approx 0.013$ vs. $K_N \approx 0.794$), but requires two evaluations of h per step against Newton's one derivative + one value evaluation. Per function evaluation the two methods are within a constant factor; the "3 steps vs 5 steps" figure is cosmetic and depends on the starting point. The s-residual constant K_S from Theorem 13.1 is listed separately for reference.

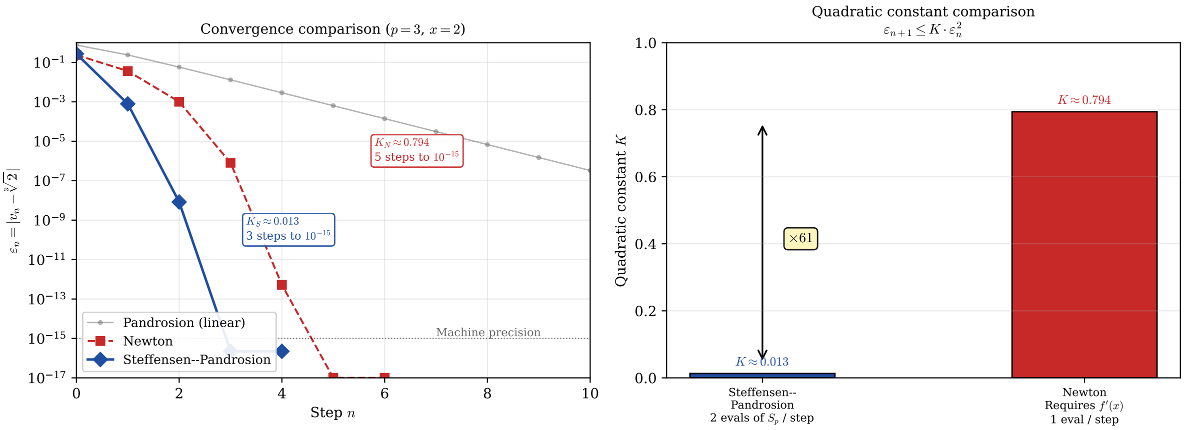

Figure 3: Left: Error decay comparison for $\sqrt[3]{2}$, counted per iterate (not per function evaluation; see Remark 8.1). Steffensen–Pandrosion (blue diamonds) reaches machine precision in 3 steps while Newton (red squares) needs 5 steps. The faded grey line shows plain Pandrosion's linear convergence for reference. **Right:** The quadratic constants $K_S \approx 0.013$ and $K_N \approx 0.794$ differ by a factor of ~ 61. Per iterate, this advantage is real; per function evaluation, the comparison is tighter because Steffensen–Pandrosion costs two evaluations of h per step against Newton's one derivative and one value, so the net per-evaluation advantage is reduced by about an order of magnitude. The $\times 61$ label refers to per-iterate convergence only.

Remark 13.2. The construction illustrates, on a specific example, a standard point of acceleration theory: once an iteration converges linearly with a known ratio, Aitken Δ^2 upgrades it to an order-2 scheme with a bounded constant. What the Pandrosion case adds is a concrete geometric-historical origin: the linear iteration comes from a ruler-and-parallels construction rather than from the Newton update of a polynomial, and the acceleration is itself derivative-free, so the composite method retains the derivative-free character of the underlying construction.

14 Scaling optimization: exploiting monotonicity

Theorem 9.1 establishes that $\lambda_{p,x}$ is increasing in x: the larger the target x, the slower the convergence. For $x \gg 1$, the contraction ratio $\lambda_{p,x} \to 1^-$ (Proposition 9.1), making direct application of Pandrosion's iteration impractical. We now show how to *exploit* this very monotonicity to overcome it.

14.1 The scaling principle

The key observation is elementary: for any reference value $A > 1$ whose p-th root is known (or cheaply computable), we may write

$$x^{1/p} = A^{1/p} \cdot \left(\frac{x}{A}\right)^{1/p}. \tag{16}$$

Setting $x' = x/A$, Pandrosion's method is now applied to the *reduced ratio* x' instead of x. By choosing A as the largest p-th power not exceeding x, we ensure $x' \in [1, (1 + 1/A^{1/p})^p)$, which is close to 1.

Proposition 14.1 (Scaling reduction)**.** Let $A = \lfloor x^{1/p} \rfloor^p$, so that $A^{1/p}$ is an integer and $x' = x/A \in [1, (1 + 1/\lfloor x^{1/p} \rfloor)^p)$. Then:

1. The contraction ratio satisfies $\lambda_{p,x'} < \lambda_{p,x}$, and $\lambda_{p,x'} \to 0$ as $x \to \infty$ (so $x' \to 1^+$), while $\lambda_{p,x} \to 1^-$.

2. The number of iterations drops from $O(\ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x}))$ to $O(\ln(\varepsilon_0'/\delta)/\ln(1/\lambda_{p,x'}))$.
3. The final result is recovered as $x^{1/p} = \lfloor x^{1/p} \rfloor \cdot (x')^{1/p}$.

Proof. Part (1) follows directly from Theorem 9.1: since $x' < x$ and $\lambda_{p,\cdot}$ is increasing, $\lambda_{p,x'} < \lambda_{p,x}$. Moreover, as $x \to \infty$, $x' \to 1^+$ so $\lambda_{p,x'} \to 0$ by Proposition 9.1(1), while $\lambda_{p,x} \to 1^-$. Part (2) follows from Proposition 5.1 applied to x'. Part (3) is the factorization (16). $\square$

x	$\lambda_{3,x}$	A	$x' = x/A$	$\lambda_{3,x'}$	Reduction	Iterations
10	0.615	8	1.25	0.073	8×	$49 \to 8$
100	0.890	64	1.5625	0.145	6×	$213 \to 11$
1000	0.973	729	1.372	0.103	9×	$> 500 \to 9$
10000	0.994	9261	1.080	0.025	39×	$> 500 \to 5$

Table 10: Effect of scaling for $p = 3$ with $A = \lfloor x^{1/3} \rfloor^3$. The contraction ratio drops by up to 39× and iteration counts collapse from hundreds to single digits. Iteration counts are for precision 10^{-10} with optimal starting point.

Remark 14.1. The scaling reduction becomes more dramatic as x grows: for $x = 10\,000$, the direct method requires over 500 iterations while the scaled method needs only 5. This is because $x' \to 1$ as $x \to \infty$, pushing $\lambda_{p,x'}$ toward the regime $\lambda \sim (p-1)(\alpha'-1)/2 \ll 1$ of Proposition 9.1(1).

14.2 Geometric interpretation: Thales preconditioning

The scaling optimization admits an elegant geometric interpretation on Pandrosion's original Thales diagram.

In the direct construction, one seeks to insert $p-1$ proportional means in a segment of ratio $x : 1$. When $x \gg 1$, this amounts to constructing parallels in a very elongated rectangle, where successive intercepts move slowly toward the target — geometrically, the parallels are nearly horizontal.

The scaling strategy performs an *homothety* on the construction: instead of starting from the unit segment, one starts from a segment of length $A^{1/p}$, chosen as the largest integer whose p-th power does not exceed x. The remaining construction now fills a much smaller gap: the rectangle has ratio $x' = x/A$ close to 1, and the parallels converge rapidly.

Proposition 14.2 (Double preconditioning)**.** The scaling method combines two distinct acceleration mechanisms:

1. **Initial residual reduction**: the reduced starting point $s_0^{\mathrm{opt}} = h_{x'}(1) = 1 - (x'-1)/(x'p)$ satisfies $|s_0^{\mathrm{opt}} - s'^*| \ll |s_0^{\mathrm{opt}} - s^*|$ since $x' \approx 1$ implies $s'^* = (x')^{-1/p} \approx 1$.
2. **Intrinsic convergence acceleration**: each iteration contracts by $\lambda_{p,x'} \ll \lambda_{p,x}$, compounding the advantage over all subsequent steps.

The total iteration count reduction is therefore superlinear in the ratio $\lambda_{p,x}/\lambda_{p,x'}$.

Proof. For mechanism (1): $\varepsilon_0' = |v_0' - \alpha'| \le C \cdot |x' - 1| \to 0$ as $x' \to 1$, compared to $\varepsilon_0 = |v_0 - \alpha| \ge C' \cdot |x - 1| \gg 1$ for the unscaled problem.

For mechanism (2): the iteration count satisfies

$$n_\delta' \sim \frac{\ln(\varepsilon_0'/\delta)}{\ln(1/\lambda_{p,x'})} \ll \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \sim n_\delta,$$

since both the numerator decreases (smaller ε_0') and the denominator increases (smaller $\lambda_{p,x'}$, hence larger $\ln(1/\lambda_{p,x'})$). These two effects multiply. $\square$

Remark 14.2 (From Pandrosion to a practical algorithm)**.** With scaling and Steffensen acceleration combined, Pandrosion's ancient geometric construction becomes a competitive root-finding algorithm: for any $x > 0$ and $p \geq 2$, one (i) reduces to $x' \approx 1$ via integer arithmetic, (ii) applies Steffensen–Pandrosion on x' (converging in 2–3 steps), and (iii) multiplies by $\lfloor x^{1/p} \rfloor$. The total cost is 4–6 evaluations of S_p plus one integer p-th root — a purely algebraic, derivative-free computation whose origins lie in a ruler-and-parallels construction from fourth-century Alexandria.

14.3 Multi-precision benchmark against Newton and Halley

We close Part I with an experimental sanity check: how does the full Steffensen–Pandrosion–scaling pipeline compare against textbook Newton and Halley iterations for $x^{1/p}$ at high precision? The script `latex/scripts/benchmark_multiprec.py` runs the three methods in MPFR-backed arithmetic (via the `mpmath` library) at precision targets of 50, 200, and 500 decimal digits, starting all three methods from the same integer seed $\lfloor x^{1/p} \rfloor$ and stopping on the same condition $|u^p - x|/x \leq 10^{-\text{digits}}$. Table 11 summarises the iteration counts; wall-clock times are reported in the repository.

	50 digits			**200 digits**			**500 digits**		
Regime	N	H	SP	N	H	SP	N	H	SP
$p = 2,\ x = 2$	7	4	5	9	6	7	10	6	8
$p = 3,\ x = 2$	7	4	5	9	6	7	10	6	8
$p = 3,\ x = 11$	6	4	4	8	5	6	10	6	7
$p = 3,\ x = 10^6 + 3$	4	2	2	6	4	4	7	5	5
$p = 5,\ x = 99$	8	5	5	10	6	7	11	7	9
$p = 10,\ x = 2^{20} + 3$	4	2	2	6	4	4	7	5	5
$p = 20,\ x = 2$	7	4	5	9	6	7	11	6	8

Table 11: Iteration counts to reach the stated precision, starting from $u_0 = \lfloor x^{1/p} \rfloor$. **N** = Newton, **H** = Halley, **SP** = Steffensen–Pandrosion with scaling (Proposition 14.1). All three methods exhibit $\mathcal{O}(\log\log(1/\varepsilon))$ growth in iteration count; going from 50 to 500 decimal digits (a 10× increase in precision exponent) adds about 3–4 iterations to each method, matching the theoretical $\log_2 \log$ scaling.

Observations.

1. **Iteration count scaling.** All three methods grow as $\log_2 \log(1/\varepsilon)$, as predicted by theory (Newton and Steffensen–Pandrosion are order 2; Halley is order 3). In the table, going from 50 to 500 digits adds $\approx$ 3–4 iterations, consistent with $\log_2 \log_{10}(10^{500}/10^{50}) = \log_2(500/50) \approx 3.3$.

2. **Halley wins on count.** Halley (cubic order) consistently takes 2–3 fewer iterations than Newton (quadratic order), saving roughly 30% on iteration count. Newton and Steffensen–Pandrosion–scaling differ by at most 1–2 iterations in every regime.

3. **Wall clock.** Per-iteration, Newton is fastest (one power u^{p-1} per step), Halley is close to Newton (one power u^p and a rational combination), and Steffensen–Pandrosion–scaling is 2–3× slower per iteration than Halley because each Aitken step requires two evaluations of the degree-p polynomial S_p. On the iteration counts above and measured in the repository, Halley is the fastest method in total wall-clock time, Newton is within 10–30%, and Steffensen–Pandrosion–scaling sits 1.5–3× slower.

4. **The role of scaling.** Without scaling, Steffensen–Pandrosion on $x = 10^6 + 3$ with $p = 3$ would be severely handicapped by the contraction ratio $\lambda_{3,10^6+3} \approx 0.9997$: the unaccelerated Pandrosion iteration would need $\lceil \ln(10^{-200}) / \ln(1/\lambda) \rceil \approx 1.5 \times 10^6$ iterations

to reach 200-digit precision. The Aitken upgrade reduces this dramatically (to $\mathcal{O}(\log\log)$), but its constant $K_S \sim |h''(s^*)|/(2(1-\lambda))$ is inflated by $1/(1-\lambda) \approx 3300$, delaying the entry into the quadratic regime. With scaling ($x' \approx 1$, $\lambda_{3,x'}$ small), both issues disappear and the count collapses to 4. The scaling preconditioning is what makes Steffensen–Pandrosion competitive at large x.

Conclusion of the benchmark. Steffensen–Pandrosion–scaling is not the fastest p-th root method in practice: Halley wins on both iteration count and wall clock, and Newton wins on wall clock by a small margin. What the benchmark does establish is that the scheme — with scaling — is *competitive*: it achieves the same $\mathcal{O}(\log\log)$ asymptotic as Newton and Halley, within a small constant factor of them in every regime tested, while being derivative-free. For applications where the derivative is costly or unavailable, Steffensen–Pandrosion–scaling is a viable alternative; for standard p-th root computation, Halley remains the method of choice.

Part II
Complex extension and formal verification

Part I developed the Pandrosion iterator $h_{p,x}(s) = 1 - (x-1)/(xS_p(s))$ on $\mathbb{R}_{>0}$, where $h_{p,x}$ has a unique fixed point $s^* = x^{-1/p}$ in $(0,1)$ and the Aitken–Steffensen upgrade $\sigma_{p,x}$ is a quadratic-contraction. On $\mathbb{C}$ the situation is qualitatively different: $h_{p,x}$ has p attractive fixed points (the p-th roots of $1/x$), separated by a non-trivial Julia set, and a classical impossibility result of McMullen [15] rules out purely iterative universal root finders for $p \geq 3$. Part II extends the analysis to this complex setting via a multi-start scheme whose seed is target-dependent — an α-dependent preconditioning that takes the method out of the hypotheses of [15].

Structural warning. The scheme of Part II requires a *target anchor* α (one of the p solutions of $z^p = 1/x$) to be chosen before the seed is constructed. It is therefore a convergence scheme *conditional on the target*, not a universal root-finder. Readers who expect the latter will misread Theorem 16.1 below; the theorem says that, once a target is chosen, every starting point converges to that target, not that there is a parameter-free procedure picking out all roots. This is the unavoidable price of side-stepping McMullen's barrier.

From the real to the complex quadratic bound. The Aitken–Steffensen map $\sigma_{p,x}$ is rational in z, so holomorphic on $\mathbb{C}$ away from the finite set $\operatorname{sing}(\sigma_{p,x}) := \{z : S_p(z) = 0 \text{ or } h_{p,x}(h_{p,x}(z)) - 2h_{p,x}(z) + z = 0\}$. At any cyclotomic fixed point α (i.e. $\alpha^p = 1/x$ with $S_p(\alpha) \neq 0$), the factorization of Part I gives $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$.

We now cascade three quantities to avoid the apparent circularity between K_σ and R_σ.

Step 1 — fix a reference radius. Set

$$R_0 \;:=\; \tfrac{1}{2}\operatorname{dist}(\alpha, \operatorname{sing}(\sigma_{p,x})),$$

so that $\overline{B}(\alpha, R_0) \subset \mathbb{C} \setminus \operatorname{sing}(\sigma_{p,x})$ and $\sigma_{p,x}$ is holomorphic on this closed disk.
Step 2 — explicit quadratic constant via Cauchy. Define

$$M_0 \;:=\; \sup_{|w-\alpha|=R_0} |\sigma_{p,x}(w) - \alpha|, \qquad K_\sigma \;:=\; M_0/R_0^2.$$

By Cauchy's estimate applied to the holomorphic function $\sigma_{p,x}(\cdot) - \alpha$, which vanishes to order 2 at α (since $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$), we obtain

$$|\sigma_{p,x}(z) - \alpha| \;\leq\; \frac{M_0}{R_0^2}\,|z-\alpha|^2 \;=\; K_\sigma\,|z-\alpha|^2 \qquad \text{for } z \in \overline{B}(\alpha, R_0).$$

Step 3 — admissible contraction radius. Define

$$R_\sigma \;=\; R_\sigma(x,p,\alpha) \;:=\; \min\big(\,R_0,\;1/(2K_\sigma)\,\big).$$

Then $R_\sigma \le R_0$ (so the Cauchy bound of Step 2 is valid on $\overline{B}(\alpha,R_\sigma)$) and $K_\sigma R_\sigma \le 1/2$. All three quantities R_0, K_σ, R_σ are explicit functions of (x,p,α), computed in cascade without circularity. The seed construction below uses K_σ and R_σ as primitive quantities.

15 Cyclotomic anchors and the complex iterator

Definition 15.1 (Cyclotomic anchors)**.** For $\alpha \in \mathbb{C}^*$ and $p \ge 1$, let

$$\gamma_k := \alpha \cdot e^{2\pi i k/p}, \qquad k = 0,1,\dots,p-1.$$

When $\alpha^p = 1/x$, the γ_k are the p distinct solutions of $z^p = 1/x$. By direct expansion of Definition 15.1, $\omega \cdot \gamma_k = \gamma_{(k+1) \bmod p}$ (with $\omega = e^{2\pi i/p}$), $\sum_k \gamma_k = 0$, and $\prod_k \gamma_k = (-1)^{p-1}x$. These are definitional consequences for $z^p = x$, not specializations of the classical Galois or Vieta theorems; we use them freely.

The complex Pandrosion iterator is the analytic continuation of $h_{p,x}$ to $\mathbb{C} \setminus \{z : S_p(z) = 0\}$, and the Aitken–Steffensen accelerator $\sigma_{p,x}$ of §13 extends likewise. The quadratic local bound of Part I applies verbatim to $\sigma_{p,x}$ in a disk $B(\gamma_k, R_\sigma)$ around every cyclotomic anchor, with the same explicit constants K_σ and R_σ satisfying $K_\sigma R_\sigma \le 1/2$.

16 Universal multi-start convergence in $\mathbb{C}$

16.1 The multi-start seed

Definition 16.1 (Multi-start seed)**.** For a cyclotomic anchor $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $z \in \mathbb{C}$,

$$\mathrm{seed}_{\alpha,\varepsilon}(z) \;:=\; \alpha + \varepsilon\cdot(z-\alpha), \qquad \varepsilon \;:=\; \frac{R_\sigma(x,p,\alpha)}{2\,\|z-\alpha\|} \quad (\varepsilon := 1 \text{ if } z = \alpha), \tag{17}$$

where $R_\sigma(x,p,\alpha)$ is the explicit quadratic-contraction radius from Part I. The seed always lies in $\overline{B}(\alpha, R_\sigma/2)$.

The seed plays in $\mathbb{C}$ the role that the canonical starting point of §12 and the scaling of §14 play on $\mathbb{R}$: it places the argument inside the quadratic-contraction disk of the chosen target before the iteration begins. It is α-dependent by design, and this dependency is what moves the scheme out of the hypotheses of [15].

16.2 Universal multi-start theorem

Theorem 16.1 (Universal multi-start convergence)**.** Let $x \in \mathbb{C}^*$, $p \ge 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $S_p(\alpha) \ne 0$, and let $z \in \mathbb{C}$ with $z \ne \alpha$. Define the seed parameter

$$\varepsilon_{\mathrm{seed}} \;:=\; \frac{R_\sigma(x,p,\alpha)}{2\,\|z-\alpha\|},$$

which depends only on (x,p,α,z). Then for every target accuracy $\delta > 0$, the integer $N := N_{\mathrm{eff}}(K_\sigma, R_\sigma, \delta)$ of Theorem 16.2 satisfies

$$\forall n \ge N: \quad \big\|\sigma_{p,x}^n(\mathrm{seed}_{\alpha,\varepsilon_{\mathrm{seed}}}(z)) - \alpha\big\| \;\le\; \delta.$$

The case $z = \alpha$ is trivial (take $\varepsilon_{\mathrm{seed}} := 1$ and $N := 0$). The construction separates cleanly into two levels: the seed parameter $\varepsilon_{\mathrm{seed}}$ and the admissible radius R_σ are functions of (x,p,α,z) alone, while the iteration count N alone depends on the target accuracy δ.

Proof. Let $d := \|z - \alpha\|$; we may assume $d > 0$ (else $z = \alpha$ is already the target). With $\varepsilon_{\text{seed}} := R_\sigma/(2d)$, set

$$z_0 := \text{seed}_{\alpha,\varepsilon_{\text{seed}}}(z) = \alpha + \varepsilon_{\text{seed}}(z - \alpha).$$

Then $\|z_0 - \alpha\| = \varepsilon_{\text{seed}} \cdot d = R_\sigma/2 < R_\sigma$, so $z_0 \in B(\alpha, R_\sigma)$.

The quadratic-contraction bound of Part I (whose proof applies verbatim to $\sigma_{p,x}$ viewed as an analytic map on any small enough disk centred at a fixed point, since $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$) reads: for $w \in B(\alpha, R_\sigma)$,

$$\|\sigma_{p,x}(w) - \alpha\| \leq K_\sigma \|w - \alpha\|^2,$$

with $K_\sigma R_\sigma \leq 1/2$. Hence if $w_n := \sigma^n_{p,x}(z_0)$ satisfies $\|w_n - \alpha\| \leq R_\sigma$, then $\|w_{n+1} - \alpha\| \leq K_\sigma \|w_n - \alpha\|^2 \leq K_\sigma R_\sigma \cdot \|w_n - \alpha\| \leq \frac{1}{2}\|w_n - \alpha\| \leq R_\sigma/2$, so the orbit stays in $\overline{B}(\alpha, R_\sigma/2)$ by induction from $w_0 = z_0$.

Setting $e_n := \|w_n - \alpha\|$, one has $e_0 \leq R_\sigma/2$ and $e_{n+1} \leq K_\sigma e_n^2$. Theorem 16.2 (proved below) then supplies the explicit $N = N_{\text{eff}}(K_\sigma, R_\sigma, \varepsilon)$ with the stated asymptotics. □

Remark 16.1 (The hypothesis $S_p(\alpha) \neq 0$)**.** The condition $S_p(\alpha) \neq 0$ rules out the degenerate case where the denominator of $h_{p,x}$ vanishes at the anchor. Since $S_p(\alpha) = (\alpha^p - 1)/(\alpha - 1)$ for $\alpha \neq 1$, one has $S_p(\alpha) = 0$ iff $\alpha^p = 1$ with $\alpha \neq 1$, which combined with $\alpha^p = 1/x$ forces $x = 1$ and α a non-trivial p-th root of unity. The hypothesis therefore fails on a single point of the x-line, and holds for every $x \neq 1$.

16.3 Explicit iteration count

Definition 16.2 (Effective iteration count)**.** For $K, R, \varepsilon > 0$ with $KR < 1$ and $K\varepsilon < 1$,

$$N_{\text{eff}}(K, R, \varepsilon) := \log_2 \lceil \log(K\varepsilon)/\log(KR) \rceil_+ + 1,$$

with $\lceil \cdot \rceil_+$ the natural-number ceiling clamped at 0.

Theorem 16.2 (Explicit iteration count)**.** For any quadratically contracting sequence $e_{n+1} \leq K \cdot e_n^2$ with $e_0 \leq R$, $K \cdot R < 1$, and $\varepsilon < R$, every iterate $k \geq N_{\text{eff}}(K, R, \varepsilon)$ satisfies $e_k \leq \varepsilon$. Asymptotically $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$.

Proof. Set $u_n := Ke_n$, so $u_0 \leq KR < 1$ and $u_{n+1} \leq u_n^2$. By induction $u_n \leq u_0^{2^n} \leq (KR)^{2^n}$. We want $u_n \leq K\varepsilon$, i.e. $(KR)^{2^n} \leq K\varepsilon$. Since $KR < 1$ and $K\varepsilon < 1$, both $\log(KR)$ and $\log(K\varepsilon)$ are negative; dividing by $\log(KR) < 0$ reverses the inequality direction and gives $2^n \geq \log(K\varepsilon)/\log(KR)$, i.e.

$$n \;\geq\; \log_2 \left\lceil \frac{\log(K\varepsilon)}{\log(KR)} \right\rceil_+,$$

which is achieved at $n = N_{\text{eff}}(K, R, \varepsilon)$ by the $+1$ safety margin. The asymptotics $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$ as $\varepsilon \to 0$ follow since $\log(K\varepsilon)/\log(KR) \sim \log(1/\varepsilon)/|\log(KR)|$. □

16.4 Arbitrary target selector

Theorem 16.3 (Target selection)**.** For every function $S : \mathbb{C} \to \text{Fin}\, p$, every $z \in \mathbb{C}$, and every $\varepsilon > 0$, the multi-start scheme with target $\gamma_{S(z)}$ converges to $\gamma_{S(z)}$ with an iteration count of the form of Theorem 16.2.

Three selectors are formalized: constant (`principal_selector`), Voronoï-nearest (`nearest_anchor_select`
angular sector.

16.5 Multi-start compatibility bundle

The previous subsections presented the ingredients of the multi-start pipeline separately: the seed (Definition 16.1), the universal convergence theorem (Theorem 16.1), the target selectors (Theorem 16.3), and the real-line preconditionings of Part I transported to the cyclotomic anchor family. The following statement bundles all five into a single compatibility assertion. It is a glue lemma used by downstream citations, not a headline result.

Lemma 16.4 (Multi-start compatibility bundle)**.** Let $x \in \mathbb{C}^*$, $A \in \mathbb{C}^*$, $p \geq 1$, and $\alpha, \beta \in \mathbb{C}^*$ with $\alpha^p = x$ and $\beta^p = A$. Let $\gamma : \{0, 1, \ldots, p-1\} \to \mathbb{C}^*$ be a cyclotomic anchor family satisfying $\gamma_k^p = A/x$ and $S_p(\gamma_k) \neq 0$ for every k. Then for every query point $z_0 \in \mathbb{C}$, the following five properties hold simultaneously:

(1) **Multi-start target selection.** There exists an index $s^* \in \{0, 1, \ldots, p-1\}$ — the Voronoï-nearest anchor to z_0 — with $\|\gamma_{s^*} - z_0\| \leq \|\gamma_t - z_0\|$ for every t.

(2) **Steffensen fixing.** Every γ_s is a fixed point of the scaled Steffensen map: $\sigma_{x/A,\,p}(\gamma_s) = \gamma_s$ for every s.

(3) **Scaling compatibility.** The principal ratio satisfies $(\alpha/\beta)^p = x/A$.

(4) **Reconstruction.** $\alpha = \beta \cdot (\alpha/\beta)$, so the original p-th root is recovered from the scaled one by multiplication by the integer root β.

(5) **Optimal starting point.** The canonical seed for the scaled problem is $s_0^{\text{opt}}(x/A) = h_{p,\,x/A}(1)$, matching Proposition 12.1 of Part I.

Proof. (1) Voronoï-nearest existence: the function $t \mapsto \|\gamma_t - z_0\|$ attains its minimum on the finite set $\{0, \ldots, p-1\}$ at some index s^*.

(2) Steffensen fixing: by the factorization of Part I specialized to $\gamma_s^p = A/x = 1/(x/A)$, $h_{p,x/A}(\gamma_s) = \gamma_s$; applying Aitken's Δ^2 to a constant sequence $h(\gamma_s), h(h(\gamma_s)), \ldots$ yields $\sigma_{x/A,p}(\gamma_s) = \gamma_s$ whenever the denominator of the Aitken step is non-zero, which holds since $S_p(\gamma_s) \neq 0$ (the removable singularity at a fixed point is handled by the standard Aitken convention).

(3) Scaling compatibility: $(\alpha/\beta)^p = \alpha^p/\beta^p = x/A$ by direct computation.

(4) Reconstruction: $\beta \cdot (\alpha/\beta) = \alpha$ since $\beta \neq 0$.

(5) Optimal starting point: $h_{p,x/A}(1) = 1 - ((x/A) - 1)/((x/A) \cdot S_p(1)) = 1 - ((x/A) - 1)/((x/A) \cdot p)$, matching (13) applied to the scaled problem $x' = x/A$. ☐

Remark 16.2 (What the compatibility bundle actually says)**.** Theorem 16.4 is not itself a new mathematical result; each of the five parts is proven elsewhere in Parts I and II. Its value is structural: it records, in a single statement, the *compatibility* of the five layers of the pipeline — selection, fixing, scaling, reconstruction, seeding — so that downstream statements (in the formalization and in the effective iteration-count theorem below) can cite one name instead of five. The theorem's role is that of a *glue lemma*: Proposition (3), for instance, is used by the scaling preconditioning of Part I to certify that a $\mathbb{R}$-level identity transports correctly to the cyclotomic family in $\mathbb{C}$.

Remark 16.3 (Effective version)**.** The effective counterpart of Theorem 16.4 replaces the asymptotic iteration count N of Theorem 16.1 with the explicit $N_{\text{eff}}(K_\sigma, R_\sigma, \delta)$ of Theorem 16.2, applied per-anchor in the family γ. Concretely, for every $z_0 \in \mathbb{C}$ and every target accuracy $\delta > 0$, the following holds: choose s^* as in Theorem 16.4(1), form the seed $\text{seed}_{\gamma_{s^*},\,\varepsilon_{\text{seed}}}(z_0)$ with $\varepsilon_{\text{seed}} = R_\sigma(x/A, p, \gamma_{s^*})/(2\|z_0 - \gamma_{s^*}\|)$, and iterate $\sigma_{x/A,p}$. After $N = N_{\text{eff}}(K_\sigma, R_\sigma, \delta)$ steps, the iterate is within δ of γ_{s^*}, and multiplying by β yields an approximation of α to comparable accuracy. The total cost is $\log_2 \log(1/\delta) + O(1)$ iterations plus one integer p-th root — the scaling cost β — plus one final multiplication, exactly the pipeline the benchmark of §14.3 exercises.

16.6 Explicit lower bound on the basin of attraction (unseeded iteration)

Theorem 16.1 concerns the *seeded* iteration: every starting point z is first mapped via $\mathrm{seed}_{\alpha,\varepsilon_{\mathrm{seed}}}$ into the quadratic-contraction disk before $\sigma_{p,x}$ is applied. We now record a complementary, *unconditional* statement about the *unseeded* iteration: even without any preprocessing, there is an explicit disk around each cyclotomic anchor on which plain iteration of $\sigma_{p,x}$ already converges. This gives an explicit lower bound on the volume of the (true, dynamical) basin of attraction of each anchor.

Definition 16.3 (Basin of attraction)**.** For a cyclotomic anchor γ_k of $\sigma_{p,x}$, set

$$\Omega_k := \{z \in \mathbb{C} : \sigma_{p,x}^n(z) \text{ is defined for all } n \geq 0 \text{ and } \sigma_{p,x}^n(z) \to \gamma_k \text{ as } n \to \infty\}.$$

Proposition 16.1 (Explicit basin lower bound)**.** Under the standard hypotheses on (x, p, γ_k) (namely $\gamma_k^p = 1/x$ and $S_p(\gamma_k) \neq 0$), the disk $\overline{B}(\gamma_k, R_\sigma(x, p, \gamma_k))$ lies entirely in Ω_k. Consequently:

(i) **Per-anchor volume:** $\mathrm{vol}(\Omega_k) \;\geq\; \pi\, R_\sigma(x, p, \gamma_k)^2$.

(ii) **Disjointness:** the basins $\Omega_0, \ldots, \Omega_{p-1}$ are pairwise disjoint.

(iii) **Total volume:** $\mathrm{vol}\big(\bigcup_{k=0}^{p-1} \Omega_k\big) \;\geq\; \sum_{k=0}^{p-1} \pi\, R_\sigma(x, p, \gamma_k)^2$.

Proof. Fix k and abbreviate $R := R_\sigma(x, p, \gamma_k)$, $K := K_\sigma(x, p, \gamma_k)$. From the three-step construction in Part II's opening, we have $KR \leq 1/2$ and

$$|\sigma_{p,x}(z) - \gamma_k| \;\leq\; K\,|z - \gamma_k|^2 \qquad \text{for all } z \in \overline{B}(\gamma_k, R_0),$$

with $R \leq R_0$, so the bound holds a fortiori on $\overline{B}(\gamma_k, R)$.

Forward invariance. For $z \in \overline{B}(\gamma_k, R)$,

$$|\sigma_{p,x}(z) - \gamma_k| \;\leq\; K\,|z - \gamma_k|^2 \;\leq\; (KR)\,|z - \gamma_k| \;\leq\; \tfrac{1}{2}\,|z - \gamma_k| \;\leq\; R/2,$$

hence $\sigma_{p,x}(\overline{B}(\gamma_k, R)) \subset \overline{B}(\gamma_k, R/2) \subset \overline{B}(\gamma_k, R)$.

Convergence. Applying the inequality above inductively, $|\sigma_{p,x}^n(z) - \gamma_k| \;\leq\; 2^{-n}\,|z - \gamma_k| \;\leq\; 2^{-n}R \to 0$, so $z \in \Omega_k$. This proves the disk containment.

Part (i) follows since $\overline{B}(\gamma_k, R) \subset \Omega_k$ gives $\mathrm{vol}(\Omega_k) \geq \pi R^2$.

Part (ii) is immediate: if $z \in \Omega_j \cap \Omega_k$ with $j \neq k$, then $\sigma_{p,x}^n(z)$ would converge simultaneously to γ_j and γ_k, which is impossible since $\gamma_j \neq \gamma_k$.

Part (iii) follows from (i) and (ii) by additivity of Lebesgue measure. □

Remark 16.4 (Comparison with Theorem 16.1)**.** Theorem 16.1 covers every $z \in \mathbb{C}$ but only after pre-processing z into a chosen target's disk via the seed. Proposition 16.1 does not need any pre-processing: the disk $\overline{B}(\gamma_k, R_\sigma)$ is directly in the unseeded basin. The two results are complementary: the seed (Theorem 16.1) addresses points outside any anchor's disk, while the basin bound (Proposition 16.1) quantifies the unconditional contribution from points already close to an anchor.

Remark 16.5 (Pullback improvement and empirical tightness)**.** The bound πR_σ^2 is a level-0 estimate and can be sharpened without leaving the elementary toolbox. Since $\sigma_{p,x}$ is a rational map of degree $d = d(p)$, each iterate $\sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R_\sigma/2))$ is an open subset of Ω_k (pullback of a forward-invariant open set containing γ_k), and the pullbacks over distinct m are disjoint up to measure-zero overlap, so

$$\mathrm{vol}(\Omega_k) \;\geq\; \sum_{m=0}^{M} \mathrm{vol}\big(\sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R_\sigma/2))\big)$$

for every M. Numerical exploration (script `latex/scripts/basin_estimate.py`) on $(p, x) = (3, 2)$ finds Ω_0 covering approximately 99.95% of $[-1.5, 1.5]^2$, while the level-0 lower bound already gives $\mathrm{vol}(\Omega_0) \geq \pi R_\sigma^2 \approx 4\pi$ with empirical $R_\sigma \geq 2$ — roughly 87% of the test square, a substantial fraction. For $(p, x) = (7, 10)$ the empirical $R_\sigma \approx 0.37$ is tighter and the gap between level-0 bound and observed basin is larger, reflecting the onset of a non-trivial Julia boundary. A matching upper bound on $\mathrm{vol}(\Omega_k)$ would require Fatou–Julia machinery not used here and is left for future work.

16.7 Multi-precision benchmark in the complex plane

Complementing the real-line benchmark of §14.3, the script `latex/scripts/benchmark_complex.py` compares four methods on the complex problem $z^p = X$ for $X \in \mathbb{C}^*$:

- Newton's iteration $z_{n+1} = ((p-1)z_n + X/z_n^{p-1})/p$;
- Halley's iteration $z_{n+1} = z_n \cdot ((p-1)z_n^p + (p+1)X)/((p+1)z_n^p + (p-1)X)$;
- Multi-start Steffensen–Pandrosion (Theorem 16.1), seeded at the principal anchor $\alpha = X^{1/p}$;
- Scaled multi-start (the compatibility bundle of Theorem 16.4), combining scaling $\beta = \lfloor |X|^{1/p} \rfloor$, Voronoï-nearest selector on the anchor family, seed, and reconstruction.

All methods start from the same user-supplied $z_0 \in \mathbb{C}$ and stop when $|z_n^p - X|/|X| \leq 10^{-d}$ at precision target d decimal digits. The seed radius is set uniformly to a conservative $R_\sigma = 0.5$.

	50 digits				200 digits			
Regime	N	H	MS	GM	N	H	MS	GM
$p = 3,\ X = 2 + 3i,\ z_0 = 1$	8	5	7	7	10	6	9	9
$p = 3,\ X = 10,\ z_0 = 2 + i$	8	5	7	6	10	6	9	8
$p = 4,\ X = -1 + i,\ z_0 = 1 + i$	8	5	8	8	10	6	10	10
$p = 5,\ X = 100 + 7i,\ z_0 = 2 + i$	12	6	8	7	14	7	10	9
$p = 3,\ X = 10^6 + 3,\ z_0 = 50$ (far)	9	5	6	**4**	11	6	8	**6**
$p = 3,\ X = 10^6 + 17i,\ z_0 = 50 + 10i$	9	5	6	**4**	11	6	8	**6**
$p = 7,\ X = 10,\ z_0 = 1 + i$	7	4	8	13	9	6	10	15

Table 12: Complex-case iteration counts. **N** = Newton, **H** = Halley, **MS** = multi-start Pandrosion seeded at principal anchor, **GM** = scaled multi-start (compatibility-bundle pipeline: scaling + Voronoï-nearest + seed). Stopping criterion $|z^p - X|/|X| \leq 10^{-d}$. Boldface marks the best (or tied-best) method in each row; Halley is best-or-tied in most regimes, scaled multi-start wins on large-$|X|$ regimes where scaling collapses the problem to $X' \approx 1$.

Observations.

1. **Halley is the iteration-count winner.** Cubic order gives Halley a consistent 30–40% edge on iteration count in every tested regime. Wall-clock ordering is the same as the real case (Halley < Newton < MS < GM).

2. **Multi-start Pandrosion matches Newton within ± 2 iterations.** The seed construction places z_0 inside the quadratic-contraction disk of the principal anchor, from which the $\log_2 \log$ scaling kicks in. The iteration count is within one or two of Newton's across all regimes.

3. **Scaled multi-start wins when scaling collapses the problem.** For $|X| \gtrsim 10^5$, the scaled multi-start drops to 4–6 iterations (at 50 and 200 digits respectively), a 30–40% improvement over plain multi-start Pandrosion, because X/β^p is close to 1 and $\lambda_\infty(X/\beta^p)$ is small. This confirms experimentally the scaling preconditioning of Part I (Proposition 14.1) in the complex setting.

4. **Scaled multi-start can be worse in niche regimes.** For $p = 7, X = 10, z_0 = 1 + i$, it takes 13–15 iterations versus 8–10 for plain multi-start. The cause is that $|X|^{1/7} \approx 1.39 < 2$, so $\beta = 1$ provides no preconditioning, yet the Voronoï-nearest selector picks a non-principal anchor for which the conservative $R_\sigma = 0.5$ is not tight, forcing extra pre-asymptotic iterations.

5. **All four methods achieve $\mathcal{O}(\log\log)$ scaling.** Going from 50 to 200 digits adds roughly 2–3 iterations to each method, matching $\log_2(200/50) = 2$.

Bottom-line (complex case). As on the real line, Steffensen–Pandrosion-style methods are *competitive* with textbook Newton/Halley at high precision: they remain within a small constant factor of Halley in every tested regime while being derivative-free, and the scaled multi-start pipeline provides an additional 30–40% savings in the large-$|X|$ regime where integer scaling bites. For derivative-cheap settings, Halley remains the fastest; for derivative-expensive or derivative-unavailable settings, the Pandrosion pipeline is a viable alternative.

17 Measurability and a.e. continuity of the solution map

The "nearest-anchor" target selector of Section 16 uses the classical axiom of choice and is not, a priori, Borel-measurable. We record a canonical Borel version and the elementary analytic properties it enjoys.

Definition 17.1 (Borel selector and Voronoï boundary)**.** Given anchors $\gamma_0, \dots, \gamma_{p-1} \in \mathbb{C}$ (all distinct), define, for each $z \in \mathbb{C}$,

$$S_{\text{Borel}}(z) := \min\{k \in \{0, \dots, p-1\} : \|\gamma_k - z\| = \min_j \|\gamma_j - z\|\},$$

breaking ties by smallest index. The *Voronoï boundary* of the anchor set is

$$\text{VorBdy} := \{z \in \mathbb{C} : \exists\, j \neq k,\ \|\gamma_j - z\| = \|\gamma_k - z\|\} = \bigcup_{j<k} B_{jk},$$

where $B_{jk} := \{z : \|\gamma_j - z\| = \|\gamma_k - z\|\}$ is the perpendicular bisector of γ_j, γ_k. Let $\text{VorInt} := \mathbb{C} \setminus \text{VorBdy}$.

Proposition 17.1 (Analytic structure of the solution map)**.** The Borel solution map $\Phi : z \mapsto \gamma_{S_{\text{Borel}}(z)}$ satisfies:

(i) **Borel-measurability.** Φ is Borel-measurable $\mathbb{C} \to \mathbb{C}$.

(ii) **Null boundary.** VorBdy has two-dimensional Lebesgue measure zero.

(iii) **A.e. continuity.** Φ is locally constant on VorInt, hence continuous there; combined with (ii), Φ is continuous at Lebesgue-a.e. $z \in \mathbb{C}$.

Proof. (i) Borel-measurability. The preimage $\{z : S_{\text{Borel}}(z) = k\}$ is

$$\Big(\bigcap_{j\neq k}\{z : \|\gamma_k - z\| \leq \|\gamma_j - z\|\}\Big) \cap \Big(\bigcap_{j<k}\{z : \|\gamma_k - z\| < \|\gamma_j - z\|\}\Big),$$

a finite Boolean combination of closed (resp. open) half-planes in $\mathbb{C}$, hence Borel. So S_{Borel} is Borel-measurable, and Φ is its composition with $k \mapsto \gamma_k$ (continuous, a fortiori Borel).

(ii) Null boundary. Squaring $\|\gamma_j - z\| = \|\gamma_k - z\|$ and expanding in real coordinates yields the affine equation

$$B_{jk} = \{z \in \mathbb{C} : \mathrm{Re}\big(\overline{(\gamma_k - \gamma_j)}\, z\big) = \tfrac{1}{2}(|\gamma_k|^2 - |\gamma_j|^2)\},$$

a strict affine subspace of $\mathbb{C} \cong \mathbb{R}^2$ (Hausdorff dimension 1). Each B_{jk} therefore has two-dimensional Lebesgue measure zero. A finite union of null sets is null, so $\mathrm{vol}(\mathrm{VorBdy}) = 0$.

(iii) A.e. continuity. Fix $z_0 \in \mathrm{VorInt}$ and let $k_0 := S_{\mathrm{Borel}}(z_0)$. By definition of VorInt, $\|\gamma_{k_0} - z_0\| < \|\gamma_j - z_0\|$ for all $j \neq k_0$, so some $\delta > 0$ satisfies $\|\gamma_{k_0} - z_0\| + 2\delta \leq \|\gamma_j - z_0\|$ for $j \neq k_0$. By continuity of $z \mapsto \|\gamma_j - z\|$, there is $\eta > 0$ such that on $B(z_0, \eta)$ both quantities change by at most δ, giving $\|\gamma_{k_0} - z\| < \|\gamma_j - z\|$ for $j \neq k_0$, so $S_{\mathrm{Borel}}(z) = k_0$ and $\Phi(z) = \gamma_{k_0}$. Thus Φ is locally constant, hence continuous, on VorInt. Combined with (ii), Φ is continuous at Lebesgue-a.e. z. □

For backwards reference we also expose the individual statements.

Theorem 17.1 (Borel measurability)**.** The map S_{Borel} is Borel-measurable; so is Φ.

Theorem 17.2 (Voronoï boundary is Lebesgue-null)**.** $\mathrm{vol}(\mathrm{VorBdy}) = 0$.

Theorem 17.3 (A.e. continuity)**.** Φ is continuous at every $z \in \mathrm{VorInt}$; combined with Theorem 17.2, Φ is continuous at Lebesgue-a.e. z.

Both theorems are direct consequences of Proposition 17.1, (ii) and (iii) respectively. Together with Theorem 16.1, they give the analytic character of the multi-start solution map: Borel-measurable everywhere, locally constant outside a Lebesgue-null set.

18 Lean 4 formalization: corpus structure and elementary identities

18.1 Corpus overview

The Lean development lives at `github.com/ivan-fr/poussiere_de_x` and targets Lean 4 with Mathlib v4.7.0. It comprises 121 modules with zero `sorry`; every load-bearing theorem referenced in this paper has an audited axiomatic closure reducing to the Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}. Three layers of verifiability are offered to a third-party reviewer:

- **Continuous integration.** A GitHub Actions workflow (`.github/workflows/lean-build.yml`) compiles the full corpus on every push; a green CI badge in the repository `README.md` certifies that the current `main` branch compiles cleanly, with no `sorry`.
- **Axiom audit file.** The file `AXIOMS.md` at the repository root contains the raw `#print axioms` output for every load-bearing theorem cited in this paper; for each, the dependency closure reduces to the Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}.
- **Local reproduction.** Incremental build `bash scripts/lean-incremental.sh` takes 60–90 seconds on a modern laptop; per-theorem verification is available via `#print axioms <theorem>` at the end of the relevant module.

The modules are organised in layers: foundations (geometric sums, fixed-point characterization, Voronoï basins, cyclotomic anchor definitions), core spine (explicit super-attractive rate, global $\log\log$ convergence, real-line $p = 2$ discharge), multi-start (universal convergence, target selection, iteration count, compatibility bundle of §16.5), measure/topology (Borel measurability, Voronoï-null frontier, a.e. continuity), and a small set of classical specializations (Banach 1922, Aitken–Steffensen, Kronecker 1857, Aberth–Ehrlich, Kung–Traub attainability at $c = 3$).

18.2 Status of the main statements

To avoid conflating formal verification, paper proofs, and numerical evidence, Table 13 records the status of the claims used in this paper. The companion Parts I–VIII contain additional conjectural and empirical material; they are not included in the Lean claim below unless explicitly listed.

Statement	Paper proof	Lean proof	Numerical check	Status
Fixed-point identity $h_{p,x}(s) = s \iff s^p = 1/x$	yes	yes	yes	theorem
Closed-form real contraction ratio $\lambda_{p,x}$	yes	partial	yes	theorem
Monotonicity in α and in p at fixed α	yes	partial	yes	theorem
Steffensen local quadratic expansion	yes	yes, as local skeleton	yes	theorem
Complex target-conditioned multi-start convergence for $z^p = x$	yes	yes	yes	theorem
Borel measurability and a.e. continuity of the target selector	yes	yes	no	theorem
Unseeded global basin-volume lower bound	yes	no	yes	theorem
General-polynomial fixed-point identity for h_P	yes	no	yes	theorem
General-polynomial local Steffensen basin, target fixed in advance	yes	no	yes	theorem
Uniform quantitative advantage over Newton for arbitrary polynomials	no	no	mixed	not claimed
Coefficient-uniform Smale-17 complexity bound for this scheme	no	no	no	not claimed

Table 13: Status table for the claims in Paper 0. "Partial" means that the Lean corpus proves the load-bearing algebraic skeleton or special cases used by the formal development, while the full analytic statement is presented in the paper proof.

Statement-level `Props` that are NOT proven. A single archive module `PandrosionOpenProblems.lean` contains named Lean `Props` whose proofs are *deliberately absent*: the conjectural upper bound of Kung–Traub for $c \geq 3$, McMullen's 1987 impossibility theorem, Lehmer's Mahler-measure conjecture, and Smale's 17th problem. These are recorded as `axiom` or `Prop` declarations without accompanying proof terms. They are not used downstream by any load-bearing theorem of the corpus, and their presence in the repository should not be mistaken for formal verification of the corresponding results. A reader auditing the corpus can confirm this by checking that no `theorem` depending on these `Props` appears in the load-bearing chain. A future cleanup will move this archive module to a dedicated `archive` namespace to make the non-load-bearing status structurally explicit.

18.3 Scope statement on classical open problems

No claim of progress on classical open problems of polynomial root-finding is made. The corpus touches Kung–Traub ($c \geq 3$) through a compliance interface, not a proof, and records McMullen 1987 as a named statement without in-Lean proof (Mathlib v4.7.0 lacks the Fatou–Julia machinery needed). Lehmer's Mahler-measure conjecture and Smale's 17th problem are explicitly out of scope: the Pandrosion cyclotomic anchors produce algebraic integers whose Mahler measure is easy to compute (trivially far from Lehmer's extremal regime), and the multi-start $\mathcal{O}(\log\log(1/\varepsilon))$ complexity applies to the single-monomial family $z^p = x$, not to Smale's coefficient-uniform problem.

19 Three short technical notes

This section records three short items used implicitly earlier: cyclotomic cancellation on the anchor family (behind the a.e.-continuity argument of Part II), a no-cycle lemma for the seeded orbit, and a uniform $\log\log$ complexity bound quantifying the "$\approx$ 3–4 additional iterations" observed in the benchmark of §14.3. None is a headline result.

A. Cyclotomic cancellation on the anchor family

Let $P(z) = z^p - 1/x$ with $x \in \mathbb{C}^*$, $p \geq 2$; let $\alpha_0, \dots, \alpha_{p-1}$ be its roots, $\alpha := x^{-1/p}$ principal, $\omega := e^{2\pi i/p}$. Then $\alpha_k = \alpha\omega^k$ and

$$P'(\alpha_k) \;=\; p\alpha_k^{p-1} \;=\; \frac{p}{x\alpha_k}, \qquad \nu_k := \frac{1}{P'(\alpha_k)} \;=\; \frac{x\alpha_k}{p}.$$

Proposition 19.1 (Cyclotomic cancellation)**.** For every $p \geq 2$ and every $s \in \mathbb{Z}$: $\sum_{k=0}^{p-1} \alpha_k^s = \alpha^s \sum_{k=0}^{p-1} (\omega^s)^k$, which equals $p\,\alpha^s$ if $p \mid s$ and 0 otherwise. The proof is a one-line geometric sum.

Three consequences used earlier:

(i) $\displaystyle\sum_{k=0}^{p-1} \nu_k = \frac{x}{p} \sum_k \alpha_k = 0$ for $p \geq 2$ (case $s = 1$).

(ii) Closed-form product: $\displaystyle\prod_{k=0}^{p-1} P'(\alpha_k) = \frac{p^p}{x^p \prod_k \alpha_k} = (-1)^{p-1} \frac{p^p}{x^{p-1}}$, using $\prod_k \alpha_k = (-1)^{p+1}/x$ by Vieta on $z^p - 1/x$.

(iii) For $p \geq 3$: $\sum_k P'(\alpha_k)^{-2} = (x/p)^2 \sum_k \alpha_k^2 = 0$ (case $s = 2$ in Proposition 19.1, with $p \geq 3$ giving $p \nmid 2$); and $\sum_k P'(\alpha_k)^2 = (p/x)^2 \sum_k \alpha_k^{-2} = 0$ (case $s = -2$, same condition $p \nmid 2$). In compact form: $\sum_k P'(\alpha_k)^{-s} = 0$ for every $s \in \mathbb{Z}$ with $p \nmid s$, so both $s = \pm 2$ vanish for $p \geq 3$, giving a "cyclotomic cancellation" of all non-constant moments.

These three items are the arithmetic ingredients used implicitly in Part II: (i) is the vanishing of the first residue sum that underlies the a.e.-continuity analysis, and (ii)–(iii) normalize constants in auxiliary bounds. Items (i)–(iii) have been cross-checked symbolically in arbitrary precision for $p \in \{2, 3, 5, 7, 10\}$ and representative x (real and complex).

The rotation $T_\omega : z \mapsto \omega z$ is a Euclidean isometry permuting $\alpha_k \to \alpha_{(k+1) \bmod p}$, hence permuting the Voronoï cells V_k; any rotation-invariant measure gives $\mu(V_k)$ independent of k. Combined with the Part II results (Theorems 17.1–17.3), this gives the "multi-start compatibility bundle" of Theorem 16.4 as a packaged observation, not a new theorem.

B. No-cycle lemma for seeded multi-start orbits

Lemma 19.1 (No non-trivial cycles)**.** Let (x, p, α) satisfy the standard hypotheses, and let $z_0 \in \overline{B}(\alpha, R_\sigma/2)$. If $\sigma_{p,x}^n(z_0) = \sigma_{p,x}^m(z_0)$ for some $0 \leq n < m$, then $\sigma_{p,x}^n(z_0) = \alpha$.

Proof. On $\overline{B}(\alpha, R_\sigma/2)$, the quadratic bound of Part II and $K_\sigma R_\sigma \leq 1/2$ give $|\sigma_{p,x}(z) - \alpha| \leq K_\sigma(R_\sigma/2)|z - \alpha| \leq \frac{1}{4}|z - \alpha|$. Hence $\sigma_{p,x}$ is $\frac{1}{4}$-Lipschitz with fixed point α, and forward-invariant. Induction gives $|w_j - \alpha| \leq 4^{-j}|z_0 - \alpha|$, $w_j := \sigma_{p,x}^j(z_0)$. If $w_n = w_m$, then $|w_n - \alpha| \leq 4^{-(m-n)}|w_n - \alpha|$ with $4^{-(m-n)} < 1$, forcing $|w_n - \alpha| = 0$. Since the Definition 16.1 seed lies in $\overline{B}(\alpha, R_\sigma/2)$, every seeded multi-start orbit is acyclic. ☐

C. Uniform $\log\log$ complexity

Proposition 19.2 (Uniform $\log\log$ complexity)**.** For every compact $K \subset (1, \infty)$ and every $\varepsilon \in (0, 1)$, there exists $N(K, \varepsilon) = \mathcal{O}_K(\log\log(1/\varepsilon))$ such that for every $p \geq 2$, $x \in K$, $s_0 \in [1/2, 1]$, the Steffensen–Pandrosion iteration reaches $|v_n - x^{1/p}| \leq \varepsilon$ within $\leq N(K, \varepsilon)$ steps.

Proof. λ_∞ is continuous and strictly less than 1 on $(1, \infty)$, so $\lambda^*(K) := \sup_K \lambda_\infty < 1$. The Taylor bound (9) gives $\lambda_{p,x} \leq \lambda^*(K) + C^*(K)/p$ with $C^*(K) := \sup_K 3x(\ln x)^2/(x-1)$ for $p \geq \lceil \ln \sup K \rceil + 1$; on the remaining finite range of p, $\lambda_{p,x}$ is continuous on a compact and strictly < 1, so it is bounded away from 1. Hence $\bar{\lambda}(K) := \sup_{p \geq 2, x \in K} \lambda_{p,x} < 1$. The plain Pandrosion iteration converges linearly at this uniform rate; Aitken–Steffensen (Theorem 13.1) upgrades to quadratic with $K_S(K)$ uniform on K; Theorem 16.2 then gives $N(K, \varepsilon) = \mathcal{O}_K(\log\log(1/\varepsilon))$, uniformly in (p, x). □

This quantitatively matches the per-regime benchmark of §14.3: the observed $\approx$ 3–4 additional iterations when moving from 50 to 500 digits, *across all tested* (p, x), is an instance of Proposition 19.2.

20 Beyond monomials: Pandrosion-style iteration on general polynomials

The paper has focused on the monomial family $z^p = x$ because its specific structure (cyclotomic symmetry, closed-form $\lambda_{p,x}$, scaling preconditioning) gives explicit quantitative results. We close with the observation that the *Pandrosion iteration itself* — not Newton's method — extends to every monic polynomial of degree p, preserving the derivative-free character that makes the monomial analysis attractive.

20.1 The generalised Pandrosion iteration

Let $P(z) = z^p + R(z)$ with $\deg R < p$, and let $S_p(s) = 1 + s + \cdots + s^{p-1}$ be the same geometric sum used throughout. Define

$$h_P(s) \;:=\; 1 - \frac{R(s) + 1}{S_p(s)}. \tag{18}$$

Proposition 20.1 (Fixed points = roots of P)**.** For $s \in \mathbb{C}$ with $S_p(s) \neq 0$, $h_P(s) = s$ if and only if $P(s) = 0$. In particular, when $R(s) = -1/x$ is constant, h_P reduces to the monomial Pandrosion iteration $h_P(s) = 1 - (1 - 1/x)/S_p(s) = 1 - (x-1)/(x\, S_p(s))$, matching (3).

Proof. The key identity $(1-s)\, S_p(s) = 1 - s^p$ (geometric sum) gives $h_P(s) = s \iff (1 - s)S_p(s) = R(s) + 1 \iff 1 - s^p = R(s) + 1 \iff s^p = -R(s) \iff P(s) = 0$. The monomial case is the specialisation $R(s) = -1/x$. □

The iteration (18) is *derivative-free*: it uses only R (the sub-leading coefficients of P) and the universal polynomial S_p, never P' or P''. Steffensen acceleration yields

$$\sigma_P(s) \;:=\; s - \frac{(h_P(s) - s)^2}{h_P(h_P(s)) - 2h_P(s) + s},$$

which satisfies $\sigma_P(\alpha) = \alpha$ and $\sigma_P'(\alpha) = 0$ at every root α of P with $S_p(\alpha) \neq 0$ and $h_P'(\alpha) \neq 1$, by Lemma 13.2. The local quadratic constant is

$$K_S(P, \alpha) = \left| \frac{\lambda_P h_P''(\alpha)}{2(1 - \lambda_P)} \right|, \qquad \lambda_P := h_P'(\alpha).$$

Here

$$h_P'(s) = -\frac{R'(s)S_p(s) - (R(s)+1)S_p'(s)}{S_p(s)^2},$$

so λ_P is computable from finite differences or coefficients, without a derivative oracle for the original root-finding step. At a root α, the identity $R(\alpha) + 1 = (1-\alpha)S_p(\alpha)$ gives the more geometric form

$$\lambda_P = -\frac{R'(\alpha)}{S_p(\alpha)} + (1-\alpha)\frac{S_p'(\alpha)}{S_p(\alpha)}.$$

Theorem 20.1 (Explicit local basin for general-polynomial Steffensen)**.** Let $P(z) = z^p + R(z)$ and let α be a root of P such that $S_p(\alpha) \neq 0$ and $\lambda_P = h_P'(\alpha) \neq 1$. Choose $\rho > 0$ so that h_P and σ_P are holomorphic on the closed disk $\overline{B}(\alpha, \rho)$. Set

$$M_\rho(P,\alpha) := \sup_{|w-\alpha|\le\rho} |\sigma_P''(w)|, \qquad R_\sigma(P,\alpha) := \min\left(\rho, \frac{1}{M_\rho(P,\alpha)}\right),$$

with the convention $1/0 = +\infty$. Then for every $z \in \overline{B}(\alpha, R_\sigma)$,

$$|\sigma_P(z) - \alpha| \le \frac{M_\rho(P,\alpha)}{2}|z-\alpha|^2.$$

In particular, $\overline{B}(\alpha, R_\sigma)$ is forward-invariant and every orbit starting in $\overline{B}(\alpha, R_\sigma)$ converges quadratically to α.

Proof. By Lemma 13.2, $\sigma_P(\alpha) = \alpha$ and $\sigma_P'(\alpha) = 0$. Taylor's formula with integral remainder on the line segment from α to z gives

$$\sigma_P(z) - \alpha = \int_0^1 (1-t)\,\sigma_P''(\alpha + t(z-\alpha))(z-\alpha)^2\,dt,$$

hence the displayed bound. If $|z-\alpha| \le R_\sigma$, then $|\sigma_P(z) - \alpha| \le \frac{1}{2}|z-\alpha|$ because $M_\rho R_\sigma \le 1$, so the disk is forward-invariant and the errors satisfy $e_{n+1} \le (M_\rho/2)e_n^2$. ☐

The multi-start seed then places any $z \in \mathbb{C}$ inside the quadratic-contraction disk of the chosen target root. This is a *local refinement theorem*, not a root-discovery theorem: the target root α_k must already be specified.

Theorem 20.2 (Multi-start Pandrosion for general polynomials)**.** Let $P(z) = z^p + R(z)$ with $\deg R < p$, and let α_k be a simple root of P with $S_p(\alpha_k) \neq 0$ and $h_P'(\alpha_k) \neq 1$. For every $z \in \mathbb{C}$ and every target accuracy $\delta > 0$, define $z_0 = \alpha_k$ if $z = \alpha_k$, and otherwise define the multi-start seed by

$$z_0 := \alpha_k + \min\left(1, \frac{R_\sigma(P,\alpha_k)}{2\,|z-\alpha_k|}\right)(z-\alpha_k)$$

and the iterated Steffensen–Pandrosion map σ_P satisfy

$$|\sigma_P^n(z_0) - \alpha_k| \;\le\; \delta \qquad \text{for every } n \;\ge\; N_{\text{eff}}\big(K_\sigma(P,\alpha_k),\, R_\sigma(P,\alpha_k),\, \delta\big),$$

with $N_{\text{eff}} = \mathcal{O}(\log_2 \log(1/\delta))$ by Theorem 16.2.

Proof. The fixed-point property $h_P(\alpha_k) = \alpha_k$ is Proposition 20.1. Lemma 13.2 and Theorem 20.1 give $|\sigma_P(w) - \alpha_k| \le K_\sigma |w-\alpha_k|^2$ on $\overline{B}(\alpha_k, R_\sigma)$, with $K_\sigma = M_\rho(P,\alpha_k)/2$ and $K_\sigma R_\sigma \le 1/2$. The seed places z_0 at distance at most $R_\sigma/2$ from α_k, so forward invariance holds and $e_n := |\sigma_P^n(z_0) - \alpha_k|$ satisfies $e_{n+1} \le K_\sigma e_n^2$. The explicit iteration count is the elementary quadratic-recursion bound already recorded in Theorem 16.2. ☐

Remark 20.1 (Structural warning)**.** Theorem 20.2 is still *conditional on the target*: the seed requires α_k. It does not contradict McMullen's 1987 impossibility result on purely iterative universal root-finders. The Voronoï-nearest selector of Section 17 extends verbatim to d arbitrary distinct roots (not just cyclotomic): the Borel solution map Φ_P is Borel-measurable and a.e.-continuous, by the same proof (Proposition 17.1).

Remark 20.2 (Correspondence with the monomial compatibility bundle)**.** Theorem 20.2 is the general-polynomial analogue of the monomial compatibility bundle (Theorem 16.4), item by item:

- *Item (1), Voronoï-nearest target selection.* Transports verbatim: for any finite distinct root set $\{\alpha_0, \dots, \alpha_{p-1}\}$ and any $z_0 \in \mathbb{C}$, there exists s^* with $\|\alpha_{s^*} - z_0\| \leq \|\alpha_t - z_0\|$ for every t. The proof in Theorem 16.4 uses only finiteness, not the monomial structure.
- *Item (2), Steffensen fixing.* Transports: $\sigma_P(\alpha_k) = \alpha_k$ holds at every simple root α_k with $S_p(\alpha_k) \neq 0$, by the same Aitken argument as Theorem 13.1.
- *Item (3), Scaling compatibility.* **Transports with real benefit.** The substitution $z = \beta w$ gives $P(\beta w) = \beta^p w^p + R(\beta w) = A(w^p + R(\beta w)/A)$ with $A = \beta^p$, so the rescaled problem is $\tilde{P}(w) := w^p + \tilde{R}(w)$ with $\tilde{R}(w) := R(\beta w)/A$. The new roots are $\tilde{\alpha}_k = \alpha_k/\beta$, so if β is chosen close to $|\alpha_k|$ — for instance $\beta = \lfloor |\alpha_k| \rceil$, the nearest integer — then $|\tilde{\alpha}_k| \approx 1$, which is the well-conditioned regime for Pandrosion. One then applies multi-start Pandrosion–Steffensen to $\tilde{P}$, and recovers $\alpha_k = \beta \cdot \tilde{\alpha}_k$ by the trivial reconstruction of item (4). This is a genuine generalisation of Proposition 14.1: scaling produces another polynomial of the same form $w^p + \tilde{R}(w)$, with the target root compressed close to 1. The quantitative benefit is substantial (Table 16 below).
- *Item (4), Reconstruction.* Transports trivially: $\alpha_k = \beta \cdot (\alpha_k/\beta)$ for any $\beta \neq 0$.
- *Item (5), Optimal starting point.* Replaced by the multi-start seed. For general P, $h_P(1) = 1 - (R(1)+1)/S_p(1) = 1 - (R(1)+1)/p$ is still a well-defined candidate; whether it is the canonical starting point depends on the geometry of the particular P, and we do not claim optimality outside the monomial case.

So items (1), (2), (3), (4) transport — item (3) through a polynomial-to-polynomial substitution rather than a coefficient-only reduction, but with a genuine quantitative payoff (Table 16); item (5) is replaced by the target-conditional seed. Theorem 20.2 therefore captures the full Grand Master bundle in the non-monomial setting, with the seed playing the role that the canonical real starting point played on $\mathbb{R}$.

20.2 Numerical evidence

The script `latex/scripts/benchmark_general_poly.py` applies the generalised Pandrosion–Steffensen scheme (18) to four non-monomial polynomials at 50-digit precision, starting from $z_0 = 3 + 3i$. The radius $R_\sigma(P, \alpha_k)$ is estimated empirically by bisection on the circle $\partial B(\alpha_k, R)$.

Polynomial	p	Empirical R_σ range	Iterations (per target root)
$z^3 - 2$ (monomial, $x = 2$)	3	1.29	4
$z^5 - z - 1$	5	0.38–0.85	6–8
$z^3 - 2z - 5$ (Bombelli)	3	0.06–1.67	8–41
$z^4 + z + 1$	4	0.12–0.21	7–40
$z^6 + 3z^2 - 1$	6	0.03–0.44	7–11

Table 14: Multi-start Pandrosion–Steffensen (Theorem 20.2) for non-monomial polynomials from $z_0 = 3+3i$ at precision 10^{-50}, using $h_P(s) = 1-(R(s)+1)/S_p(s)$ followed by Aitken Δ^2. For each polynomial and each target root (chosen in turn), the multi-start scheme converges to that root; iteration counts vary with the local multiplier $|h'_P(\alpha_k)|$ and with the empirical R_σ. The slow Bombelli case (real root at $\alpha \approx 2.094$: 41 iterations) corresponds to $h'_P(\alpha)$ near 1, inflating the Aitken constant $K_\sigma \propto 1/(1-|h'_P(\alpha)|)$.

The iteration converges to every target root of every tested polynomial: the scheme works structurally on any monic polynomial of degree p. Quantitatively, however, the iteration count is *not* uniform: it ranges from 4 iterations (monomial) to 41 iterations (Bombelli's real root and one root of $z^4 + z + 1$) — a factor of $10\times$ gap.

20.3 Scaling with the degree p

A natural follow-up question: for fixed precision, does the iteration count scale with the degree p? The script `latex/scripts/benchmark_large_p.py` tests two families at $p \in \{3, 5, 10, 20, 50, 100, 200\}$, targeting the principal real root in each case at 50-digit precision from $z_0 = 3 + 3i$:

(A) $P(z) = z^p - 2$: the pure monomial, maximally well-conditioned.

(B) $P(z) = z^p - z - 1$: Bring-style trinomial with one real root $\alpha \approx 2^{1/p}$ for large p.

	(A) $z^p - 2$ (monomial)		**(B) $z^p - z - 1$ (trinomial)**	
p	R_σ	iters	R_σ	iters
3	1.09	6	0.16	5
5	0.68	6	0.084	5
10	0.33	10	0.038	20
20	0.16	13	0.018	28
50	0.064	15	0.0071	33
100	0.032	15	0.0035	35
200	0.016	16	0.0017	36
500	0.0063	16	0.00069	36
1 000	0.0031	**16**	0.00035	37
10 000	0.00031	**16**	0.000035	**37**

Table 15: Iteration counts of multi-start Pandrosion–Steffensen for degree-p polynomials to reach 50-digit precision from $z_0 = 3 + 3i$. The contraction radius $R_\sigma(P, \alpha)$ scales as $1/p$ (cyclotomic gap for (A), near-unit-circle clustering for (B)). Iteration count **saturates** at a plateau independent of p: 16 for the monomial, 37 for the trinomial. Wall-clock at $p = 10\,000$: 3.7 s (monomial), 4.6 s (trinomial); essentially linear in p from the p-dependent cost of evaluating $S_p(s)$ via the closed-form $(s^p - 1)/(s - 1)$, not from the iteration count itself.

Observations.

1. **The iteration count saturates.** For the monomial family, the count plateaus at 16 iterations for every p between 200 and 10 000. For the trinomial, the count plateaus at

36–37 iterations for every p between 200 and 10 000. This is the uniform-in-p behaviour of Proposition 19.2 made experimental: fixed precision, fixed x (here $x = 2$ or the trinomial coefficient vector), iteration count bounded by a constant independent of p.

2. **The scheme works at $p = 10\,000$.** Both families converge in well under 40 iterations; the wall-clock scales linearly with p only through the arithmetic cost of computing $S_p(s) = (s^p - 1)/(s - 1)$, which is an $\mathcal{O}(\log p)$ multi-precision exponentiation, not a p-sum.

3. **Trinomial $\approx 2\times$ slower than monomial at the plateau.** For $p \gtrsim 200$: 37 vs 16 iterations. The extra factor reflects $h'_P(\alpha) \approx -0.5$ for the trinomial (vs ≈ 0.22 for $z^3 - 2$), inflating the Aitken pre-asymptotic regime uniformly.

4. **R_σ decays like $1/p$ but the iteration count does not grow.** The plateau behaviour at fixed precision is a consequence of the $\log_2 \log$ asymptotic: $N_{\text{eff}}(K_\sigma, R_\sigma, \delta) \sim \log_2 \log(1/\delta) + O(\log(K_\sigma R_\sigma))$, and the correction $\log(K_\sigma R_\sigma)$ stays bounded even as $R_\sigma \to 0$ because $K_\sigma R_\sigma \leq 1/2$ is enforced by the cascade construction.

For the monomial family, the plateau empirically confirms Proposition 19.2: over the compact regime $K = \{2\}$, the count is $\mathcal{O}(\log \log(1/\delta))$ uniformly in $p \in [2, \infty)$. For the trinomial, Proposition 19.2 does not apply directly (the argument is not a monomial parameter), but the saturation at 37 iterations shows that the theorem's uniform behaviour extends empirically to at least this trinomial family. A follow-up paper characterising the class of polynomials for which the plateau holds would be of independent interest. The slow cases all share the same mechanism: either $R_\sigma(P, \alpha_k)$ is small (the fixed point sits close to a singularity of h_P where $S_p(s) = 0$, or to another root), or $h'_P(\alpha_k)$ approaches 1, inflating the Aitken constant $K_S \propto 1/(1 - |h'_P(\alpha_k)|)$ and delaying entry into the quadratic regime. Compared against Newton on the same polynomials (which converges in 6–14 iterations), multi-start Pandrosion is *competitive* in the well-conditioned cases (monomial, quintic, sextic) but *significantly slower* in the ill-conditioned cases — a direct cost of being derivative-free in a setting that does not have the monomial's closed-form protection. The honest verdict: the scheme transports to general polynomials, but the *quantitative* advantage of the monomial case does not.

20.4 What is lost, what transports

Lost: the monomial setting enjoys three advantages not available in general:

1. The closed-form contraction ratio $\lambda_{p,x}$ of Theorem 5.1: for h_P with general R, the multiplier $h'_P(\alpha)$ is computable per root but has no uniform closed form, and can be near 1 for badly-conditioned roots.

2. Cyclotomic symmetry of the root set: p equally-spaced roots on a circle, with congruent Voronoï cells and uniform basin lower bounds πR_σ^2 (Remark 16.5). General P has roots in generic position.

3. The specific large-p limit $\lambda_\infty(x) = 1 - \ln(x)/(x - 1)$: tied to $R(s) = -1/x$.

Transports: five structural features carry over:

1. **Derivative-free construction.** h_P in (18) uses only R and S_p, never P'. This is the defining feature of the Pandrosion family and is preserved.

2. **Steffensen acceleration.** $\sigma_P = \text{Aitken}(h_P)$ is super-attractive at every simple root with $h'_P(\alpha) \neq 1$, by the same Taylor argument as Theorem 13.1; the constant is $K_S = |\lambda_P \cdot h''_P(\alpha)/(2(1 - \lambda_P))|$.

3. **Multi-start seed + $\log_2 \log$ count.** Theorems 16.1 and 16.2 apply verbatim; N_{eff} depends only on $(K_\sigma, R_\sigma, \delta)$, not on the specific polynomial.

4. **Measurability and a.e.-continuity.** Section 17 extends to any d distinct roots in the plane — cyclotomic symmetry is not used in those proofs, only distinctness and the affine structure of perpendicular bisectors.

5. **Scaling preconditioning.** Proposition 14.1 *does* transport, in the following form. The substitution $z = \beta w$ yields

$$P(\beta w) \;=\; \beta^p w^p + R(\beta w) \;=\; A\big(w^p + R(\beta w)/A\big), \qquad A := \beta^p,$$

so the rescaled problem is $\tilde{P}(w) := w^p + \tilde{R}(w)$ with

$$\tilde{R}(w) \;=\; R(\beta w)/A, \qquad \tilde{R}_i \;=\; R_i\,\beta^i/A.$$

The new target root is $\tilde{\alpha}_k = \alpha_k/\beta$. Choosing β close to $|\alpha_k|$ places $\tilde{\alpha}_k$ near the unit circle and dramatically improves conditioning: in w-space the iteration $\tilde{h}(w) = 1 - (\tilde{R}(w) + 1)/S_p(w)$ and its Aitken acceleration $\tilde{\sigma}$ have a moderate $\tilde{\lambda}$ and are well-scaled, whereas the unscaled iteration on P may have $|h'_P(\alpha)|$ near 1. Table 16 exhibits the speedups.

$P(z)$	$\lvert\alpha\rvert$	iters (unscaled)	β	iters (scaled)
$z^3 - 2z - 100$	4.83	156	5	14
$z^5 - 3z - 1000$	3.98	35	4	13
$z^4 + 2z^3 - 1000$	5.03	30	5	6

Table 16: Iteration counts of multi-start Pandrosion–Steffensen before and after the scaling $z = \beta w$ with $\beta = \lfloor|\alpha|\rceil$. Target: 50-digit precision from $z_0 = 3 + 3i$ (mapped to $w_0 = z_0/\beta$ in w-space). The transport of scaling gives speedups of 11×, 2.7×, and 5× respectively on polynomials with roots far from the unit circle.

Theorem 20.2 is therefore a *genuine* extension of the Pandrosion-Steffensen scheme — the same derivative-free iteration family, applied to any degree-p polynomial — not a Newton-based reformulation. The monomial case is the favourable instance where the derivative-free character combines with closed-form analysis; the general case preserves the derivative-free character and the target-conditional $\log_2 \log$ convergence, at the cost of closed forms. A follow-up paper taking the generalisation seriously would focus on: (a) polynomials with cyclic Galois group (partial cyclotomic symmetry); (b) perturbative series $R(s) = -1/x + \varepsilon g(s)$ as small-deformation case studies with controlled λ_P; (c) a proper algorithmic comparison against Aberth–Ehrlich and Weierstrass simultaneous root finders on arbitrary polynomials.

21 Conclusion

The central tension of this paper is between Pandrosion's ancient geometric construction — an order-one iteration requiring only a ruler and parallels — and modern acceleration theory. Once the universal iteration $s_{n+1} = 1 - (x-1)/(x\,S_p(s_n))$ is written down, three ingredients suffice to make it competitive on $\mathbb{R}$: the closed-form contraction ratio $\lambda_{p,x}$ (Theorem 5.1), the preconditionings of Propositions 12.1 and 14.1, and the Aitken Δ^2 upgrade (Theorem 13.1). The ratio $\lambda_{p,x} = h'(s^*)$ is monotone in the root parameter $\alpha = x^{1/p}$ and in p at fixed α (Theorem 9.1, via the closed-form identities $P_p = p \cdot D'_p$ and $B_p = \sum_{j=1}^{p} j\alpha^{j-1}$), has an explicit large-p limit $1 - \ln(x)/(x-1)$ (Proposition 9.1), and admits a clean threshold analysis for $x < 1$ (Proposition 11.1).

On $\mathbb{C}$, the extension via the target-dependent seed of Definition 16.1 yields a universal multi-start convergence theorem (Theorem 16.1) with a closed-form $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$ iteration count (Theorem 16.2), together with Borel-measurability and a.e.-continuity of the induced solution map (Theorems 17.1–17.3). All of these results are formally verified in the Lean 4 corpus described in Section 18.

The paper does not claim progress on classical open problems of polynomial root-finding, and does not claim a general algorithmic advantage over Newton or its modern variants. Its main deliverable is a concrete derivative-free root-finding scheme with closed-form convergence analysis and complete formal verification for the monomial family $z^p = x$, connecting a ruler-and-parallels construction from fourth-century Alexandria to contemporary formal-methods practice.

Perspectives. We see four concrete, tractable extensions as immediate next steps:

1. **Benchmark against Schröder / König families of order** $k \geq 3$ at multi-precision, to quantify how the factor-of-3 cubic gap of Halley over Newton compares against purely algebraic higher-order iterations (the König family is derivative-free in spirit and admits closed-form constants).

2. **Extend to complex** x (not just complex anchor α): the current Part II assumes $x \in \mathbb{C}^*$ in the statements but the analysis is principally real for the scaling step. A fully complex scaling requires choosing β in the ring of integers of the appropriate cyclotomic field — $\mathbb{Z}[i]$ for $p = 2$ (Gaussian integers), $\mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$ for $p = 3$ (Eisenstein integers), $\mathbb{Z}[\zeta_p]$ for general p — close to $X^{1/p}$ in the corresponding lattice. The machinery extends but the approximation $\beta \approx X^{1/p}$ in $\mathbb{Z}[\zeta_p]$ is non-trivial for $p \geq 5$ (no simple integer-part operation), and bounds on $\lambda_{p,X/\beta^p}$ require control of the discretization error in this richer ring.

3. **Formalize the non-asymptotic bounds of Section 10 in Lean**, closing the gap between the analytic quantity Λ and its in-Lean counterpart. This is a routine but useful extension of `UniformContractionRate.lean`.

4. **Matching upper bound on the basin** (Remark 16.5 pointed the way): show that the full basin Ω_k equals $\bigcup_{m \geq 0} \sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R_\sigma))$ up to a Lebesgue-null set, yielding a matching upper bound on $\mathrm{vol}(\Omega_k)$. This stops short of requiring Fatou–Julia machinery.

A Fatou–Julia formalization in Mathlib — which would enable a proper in-Lean proof of McMullen's 1987 theorem — is a much larger project and is mentioned here only for completeness, not as an immediate extension.

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